

LANESKIFF · BUILD PLAN

LaneSkiff prototype build plan

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	LSK-BLD-001	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/laneskiff		
Source	docs/05-build-plan.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	First build plan; design made constructable (LSK-DDR-002)

LaneSkiff prototype build plan

Plan, not yet built. How to build the first proof-of-concept LaneSkiff, component by component, on the plywood and epoxy path. Building and testing to it is TRL 4 work. Decisions still to be made are kept in the design decisions register (docs/06-design-decisions.md), not here.

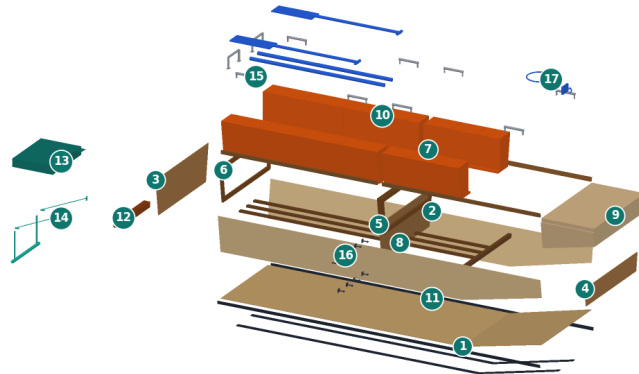
Safety: This plan builds a boat that will carry people in floodwater. Epoxy and glass work needs gloves, eye protection and a dust mask for sanding, in a ventilated space. Nobody is carried, and no boarding or swamp trial is run, until the safety stops in section 6 are passed. LaneSkiff is an open engineering reference, not certified rescue equipment, and is never used in moving water.

1. What you are building

LaneSkiff: every component, in build order

Both halves shown in place along the boat; parts pulled apart up, down, fore and aft. Plan, not yet built

- 1 Bottom panels (3)
- 2 Side panels (4)
- 3 Stern transom
- 4 Bow transom
- 5 Joint bulkheads (2)
- 6 Ring frames (3)
- 7 Inwales (4)
- 8 Stringers (6) and knuckle floor
- 9 Bow box wall, deck and foam
- 10 Bench modules (4): rail, cover, foam, straps
- 11 Rub strakes (4) and skids (4)
- 12 Hinge rail, backing block, bolts
- 13 Step float with strap hinges
- 14 Stop straps, pad eyes, kick rung
- 15 Grab and carry handles
- 16 Joint bolts and nut plates (8)
- 17 Push pole, paddles, bow eye and line



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/Laneskiff

Figure 1. Every component in build order, pulled apart. Both halves are shown in place along the boat.

A flat-bottomed skiff 3.6 m long and 1.137 m wide over its rub strakes, in two closed halves of 1.8 m that bolt together. Each half is stitched and glued from flat 6 mm plywood panels, framed with hardwood, glassed outside, and fitted with covered foam modules along both sides. The aft half carries a floating boarding step behind its transom; the forward half has a foam-filled bow box under a short deck. Fourteen components are made (the plywood panels, hardwood frames, inwales and stringers, the bench modules, the bow box, the step rail, float and kick rung, and the push pole); the rest are bought (foam sheet, HDPE strip, bolts and nut plates, hinges, webbing, handles, paddles, rope). The parts cost about USD 1,920 from the bill of materials.

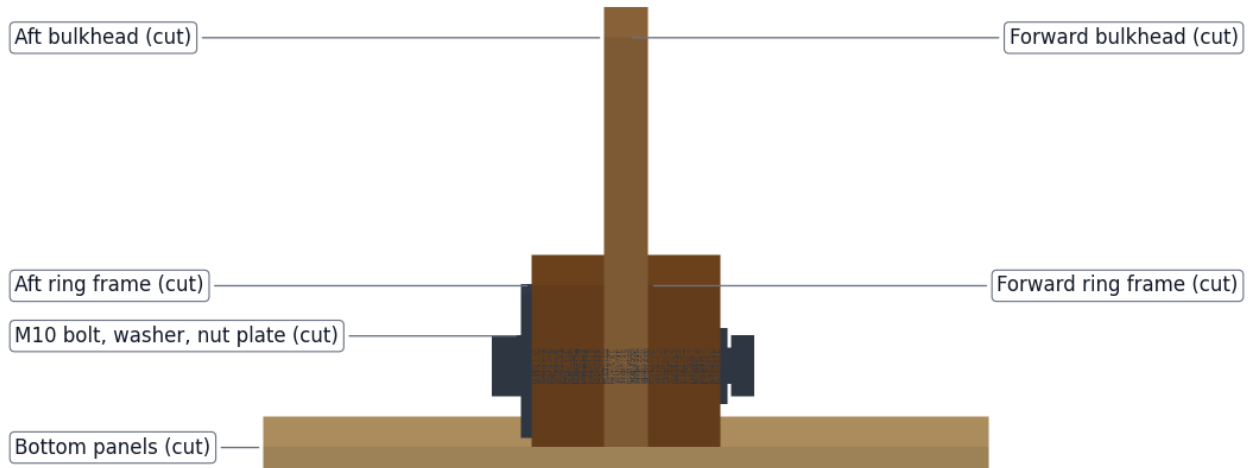
■ 2. What changed to make it buildable

Table 1. Changes from the concept.

Component	The concept had	The buildable design has	Why
Hull	A hull shape	Three flat bottom panels and four flat side panels, stitched and glued with epoxy fillets and glass tape	Every panel is cut flat from a standard sheet
The split	One mid bulkhead	Two bulkheads, one closing each half, each with a hardwood ring frame	Each half floats alone; the joint needs no seal
The joint	Bolts	Eight stainless M10 bolts from the forward side into nut plates on the aft frame, and two alignment pins	One spanner joins the halves
Side buoyancy	Blocks under benches	Foam modules in sewn tarpaulin covers, held under the inwale and strapped to a hardwood rail on the floor	Lighter than plywood boxes; the straps hold the module down when swamped
Bow buoyancy	Buoyancy in the bow	A bow box with a plywood wall and deck, filled with foam	Foam forward and high; a dry deck for the bow line
Floor	A floor	Three hardwood stringers per half and a knuckle floor	Short floor spans; the skids screw into hardwood
Stern step	A hinged buoyant step	A float on strap hinges behind a hardwood rail, a 40 mm gap, stop straps and a kick rung	Nothing can be trapped as it swings; the straps take a kneeling person
Strakes, skids, handles	Named only	HDPE strip screwed into hardwood; bought handles bolted through the inwales and transom frame; shoulder slings	Every fixing lands in hardwood
Push pole	A pole	Two 1.5 m sections with a joining sleeve	Stows inside one half

Joint 4: the halves bolted together, cut through a bolt

Bolt from the forward side into the welded nut plate; one 17 mm spanner



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

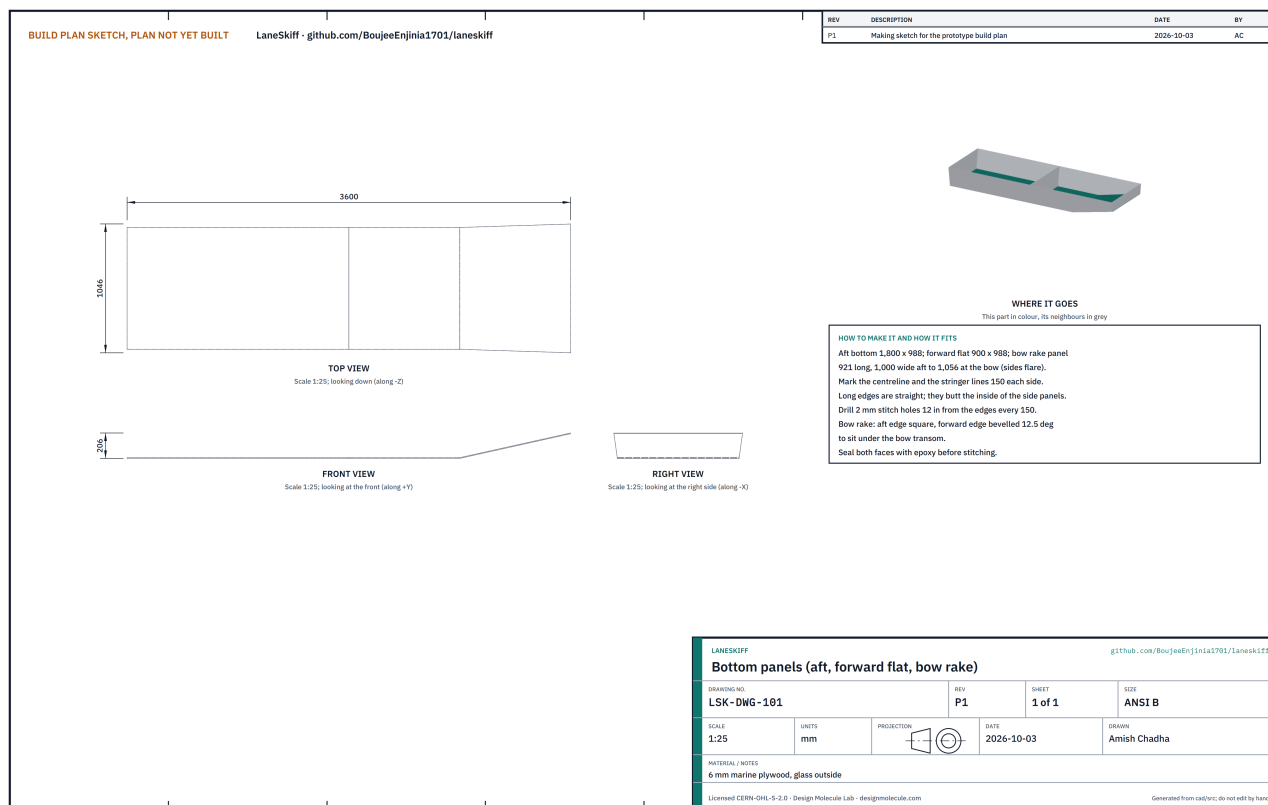
github.com/BoujeeEnjinia1701/laneskiff

Figure 2. The biggest change: two closed halves, bolted through their ring frames.

■ 3. Making the components

Make every plywood panel first, seal it with epoxy on both faces, and dry-fit before any glue. The aft half and the forward half are built the same way; differences are noted.

3.1 Bottom panels



What it is and what it is made from. Three panels of 6 mm marine plywood: the aft bottom (1,800 x 988 mm), the forward flat bottom (900 x 988 mm), and the bow rake panel (921 mm long, 1,000 mm wide at its aft edge widening to 1,056 mm at the bow, because the sides flare).

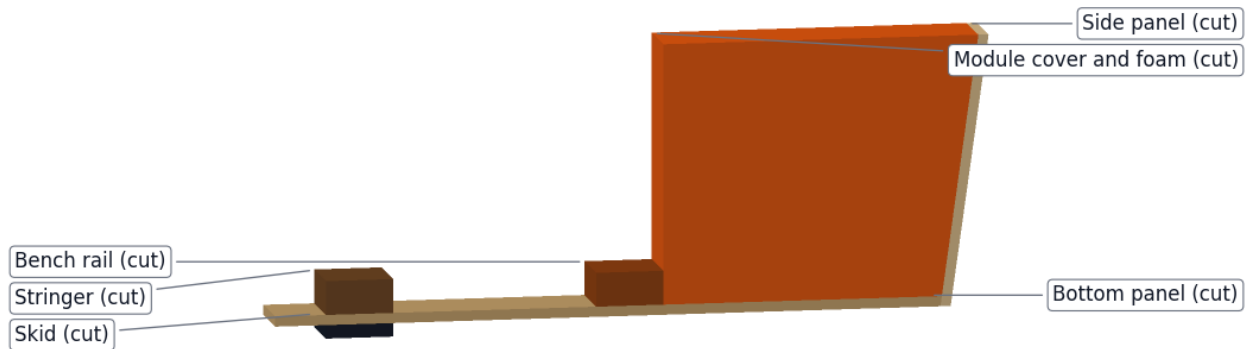
How to make it.

1. Mark the centreline and the lines of the three stringers (on the centreline and 150 mm each side) on the inside face.
2. Cut the straight long edges; they butt against the inside of the side panels.
3. Bevel the forward edge of the bow rake panel 12.5 degrees so it sits flat under the bow transom.
4. Drill 2 mm stitching holes 12 mm in from the long edges every 150 mm.
5. Seal both faces with epoxy.

How it fits the parts next to it. The side panels stand on the outside of the bottom panel's long edges and their bottom edges are flush with its underside. Inside, an epoxy fillet fills the corner; outside, 100 mm glass tape runs along the chine.

Joint 1: chine, stitched and glued

Side edge outside the bottom edge; epoxy fillet inside, 100 mm glass tape inside and out



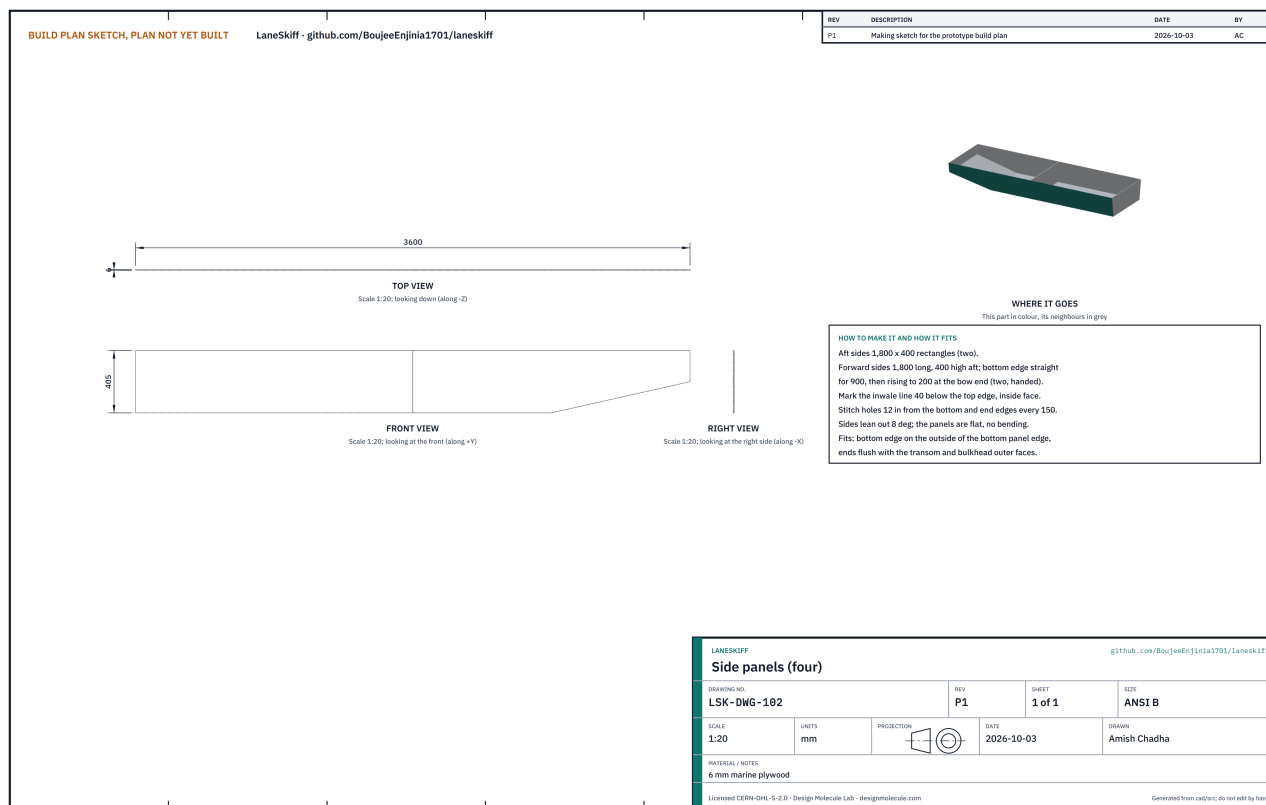
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/laneskiff

Figure 3. The chine: side panel outside the bottom edge, stringer and bench rail on the floor, skid below.

Check before moving on. Each panel within 2 mm of size; diagonals of the aft bottom equal within 3 mm.

3.2 Side panels



What it is and what it is made from. Four panels of 6 mm marine plywood, 1,800 mm long. The two aft sides are 400 mm high rectangles. The two forward sides are 400 mm high at their aft end, with the bottom edge straight for 900 mm and then rising to 200 mm at the bow end; make them as a mirrored pair.

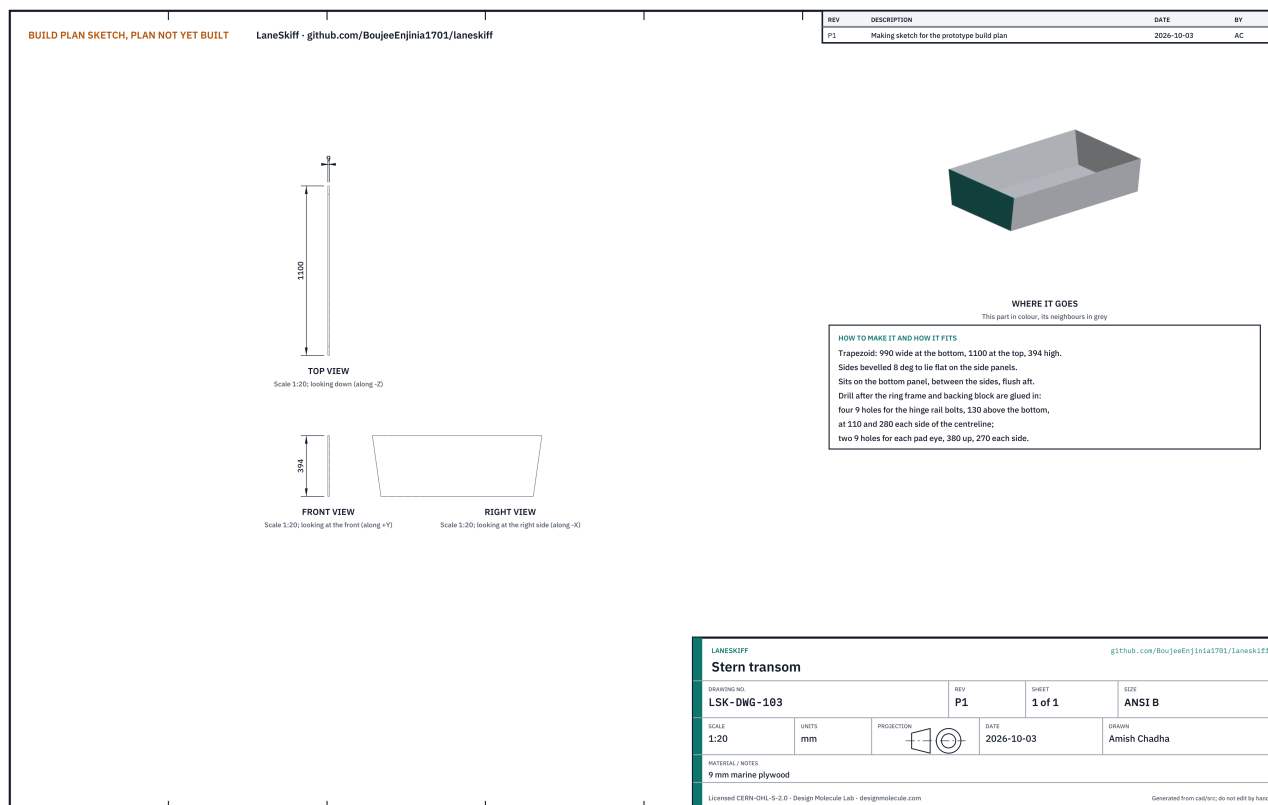
How to make it.

1. Cut both aft sides from one sheet and both forward sides from another, two to a sheet.
2. Mark the inwale line 40 mm below the top edge on the inside face.
3. Drill stitching holes 12 mm in from the bottom and end edges every 150 mm.
4. Seal both faces.

How it fits the parts next to it. The panels lean out 8 degrees from vertical. Their ends are flush with the outer faces of the transom, bulkhead and bow transom. They stay flat; there is no bending.

Check before moving on. Forward sides are a true mirrored pair: lay one on the other, edges within 2 mm.

3.3 Stern transom



What it is and what it is made from. 9 mm marine plywood, a trapezoid 394 mm high, about 988 mm wide at the bottom and 1,099 mm at the top, with both side edges bevelled 8 degrees.

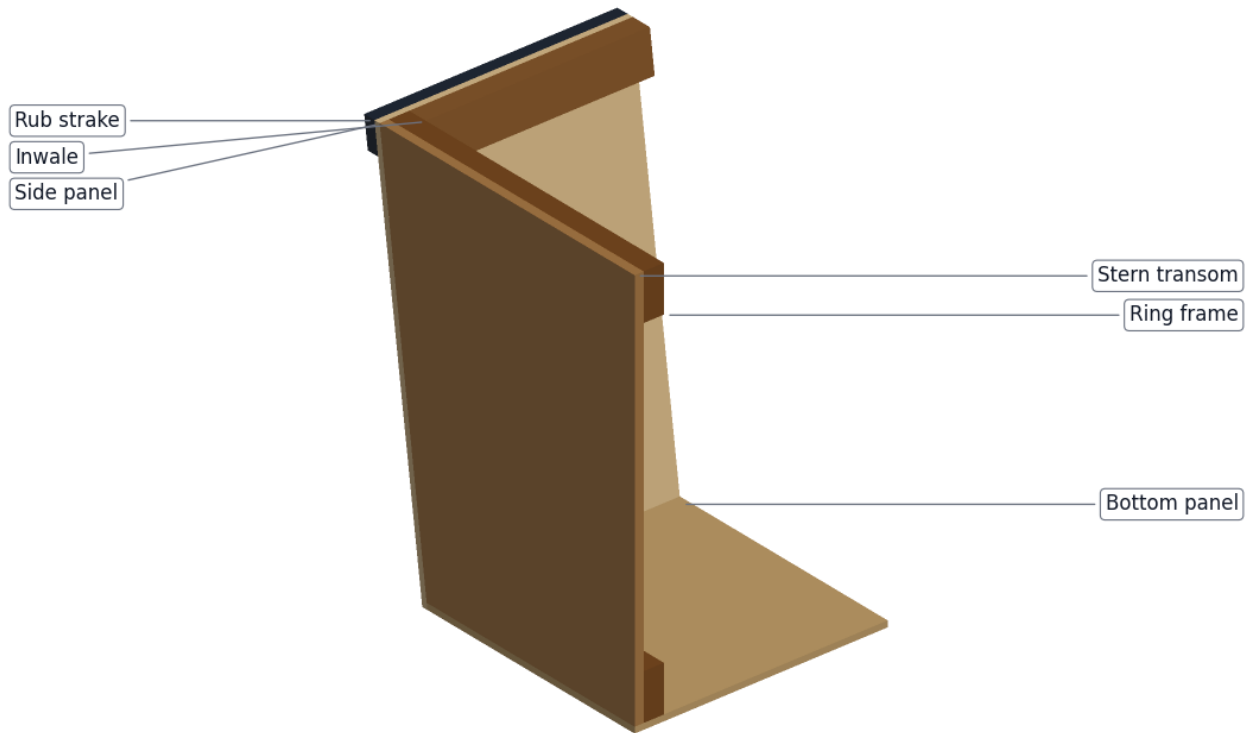
How to make it.

1. Cut the trapezoid and bevel the side edges.
2. Seal both faces.
3. After the ring frame and backing block are glued in (section 3.6 and 3.11), drill the four 9 mm holes for the hinge rail bolts, 130 mm above the bottom, at 110 mm and 280 mm each side of the centreline, and the holes for the two stop strap pad eyes, 380 mm up and 270 mm each side.

How it fits the parts next to it. It sits on the aft bottom panel, between the side panels, flush with their aft ends. Fillets and tape inside; tape outside.

Joint 2: transom corner, from inside

Transom between the sides, on the bottom; ring frame glued and screwed inside



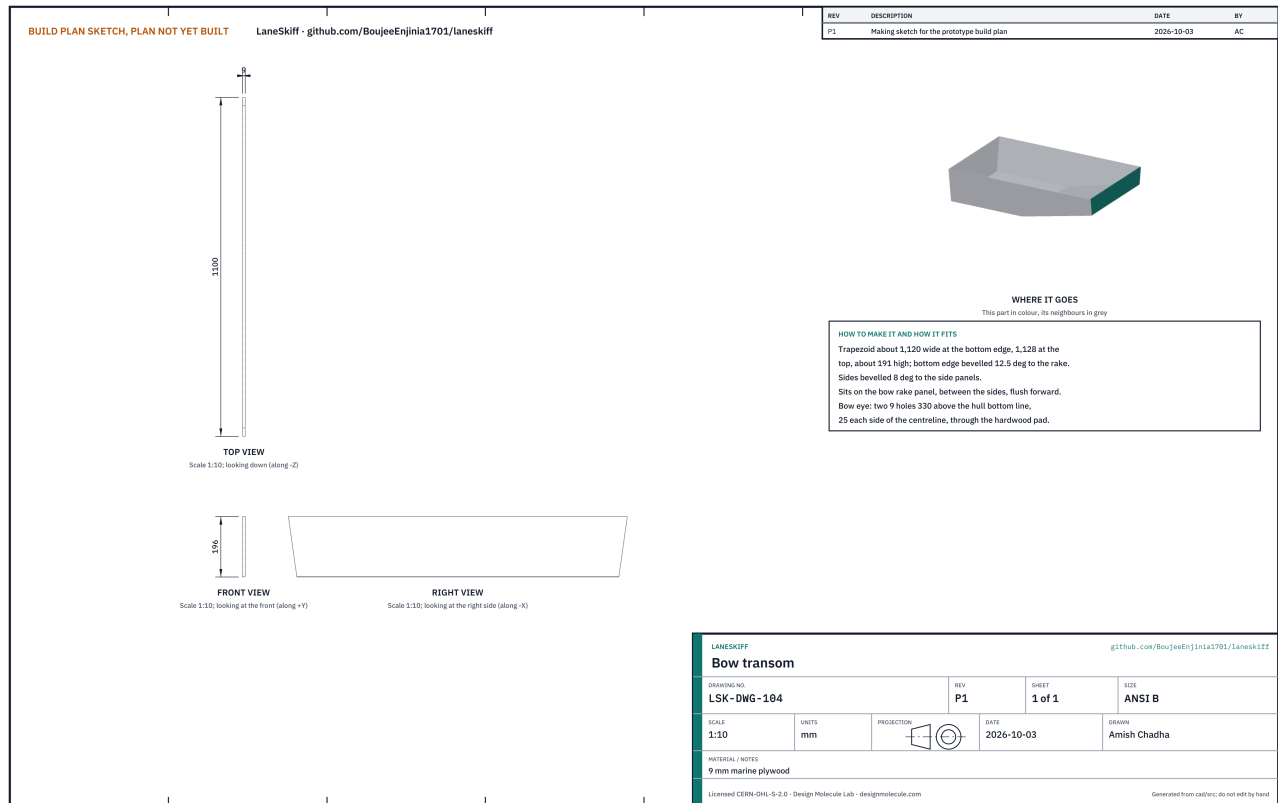
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/laneskiff

Figure 4. Transom corner: transom between the sides, ring frame inside, inwale and rub strake at the top.

Check before moving on. Square to the bottom within 1 degree.

3.4 Bow transom



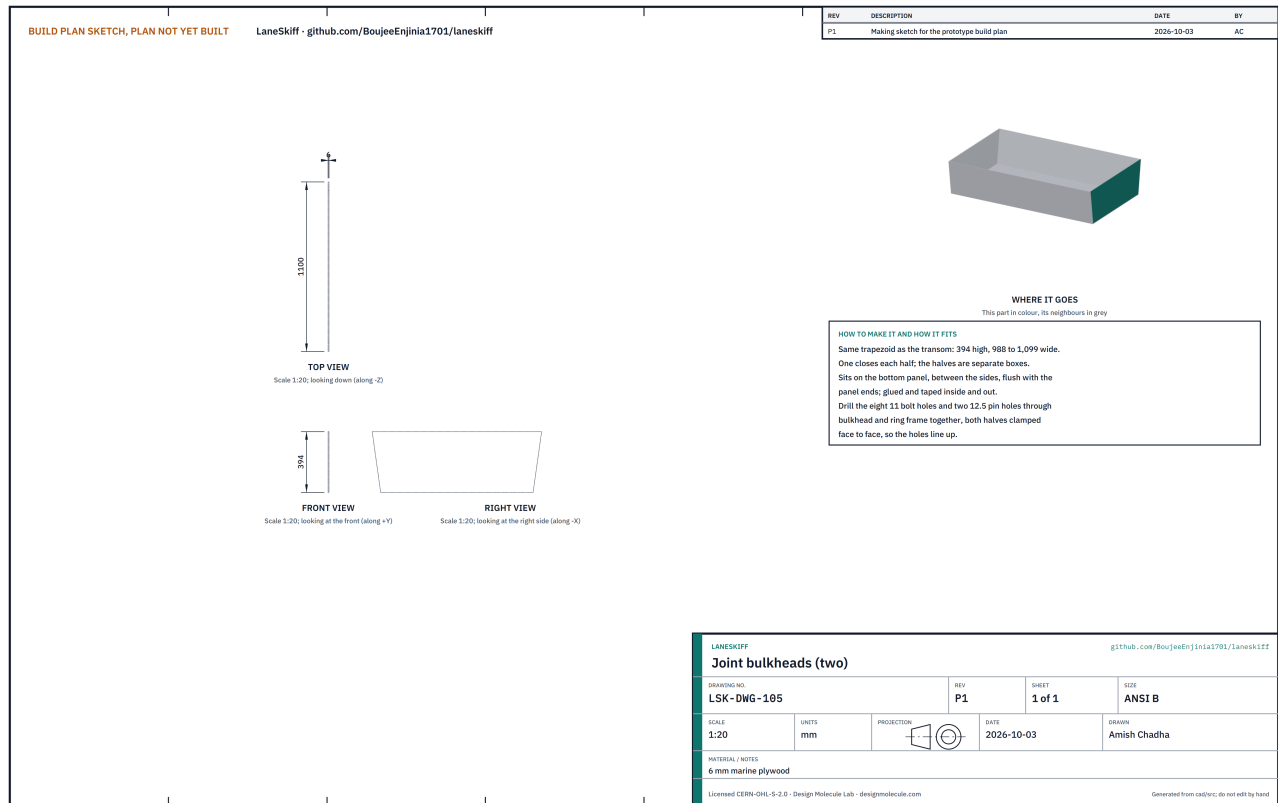
What it is and what it is made from. 9 mm marine plywood, about 1,120 mm wide at the bottom and 1,128 mm at the top, about 191 mm high, with its bottom edge bevelled 12.5 degrees to the rake and its sides bevelled 8 degrees.

How to make it. Cut and bevel; drill two 9 mm holes for the bow eye, 330 mm above the line of the hull bottom and 25 mm each side of the centreline; seal.

How it fits the parts next to it. It sits on the bow rake panel between the forward side panels, flush with their bow ends. The bow eye pad is glued to its inside (section 3.10).

Check before moving on. Top edge level with the side panels' top edges.

3.5 Joint bulkheads



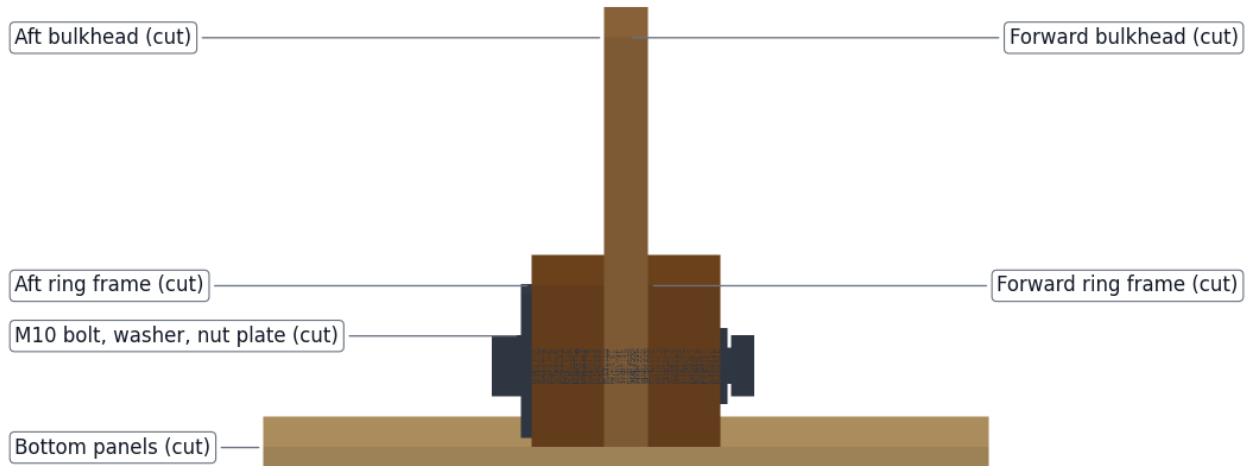
What it is and what it is made from. Two panels of 6 mm marine plywood, the same trapezoid as the stern transom.

How to make it. Cut both from one pattern; seal. Drill the bolt and pin holes later, through bulkhead and ring frame together, with both halves clamped face to face (section 3.6).

How it fits the parts next to it. Each closes its half at the joint: it sits on the bottom panel between the side panels, flush with their ends, filleted and taped inside and out. The two bulkheads touch face to face when the boat is assembled.

Joint 4: the halves bolted together, cut through a bolt

Bolt from the forward side into the welded nut plate; one 17 mm spanner



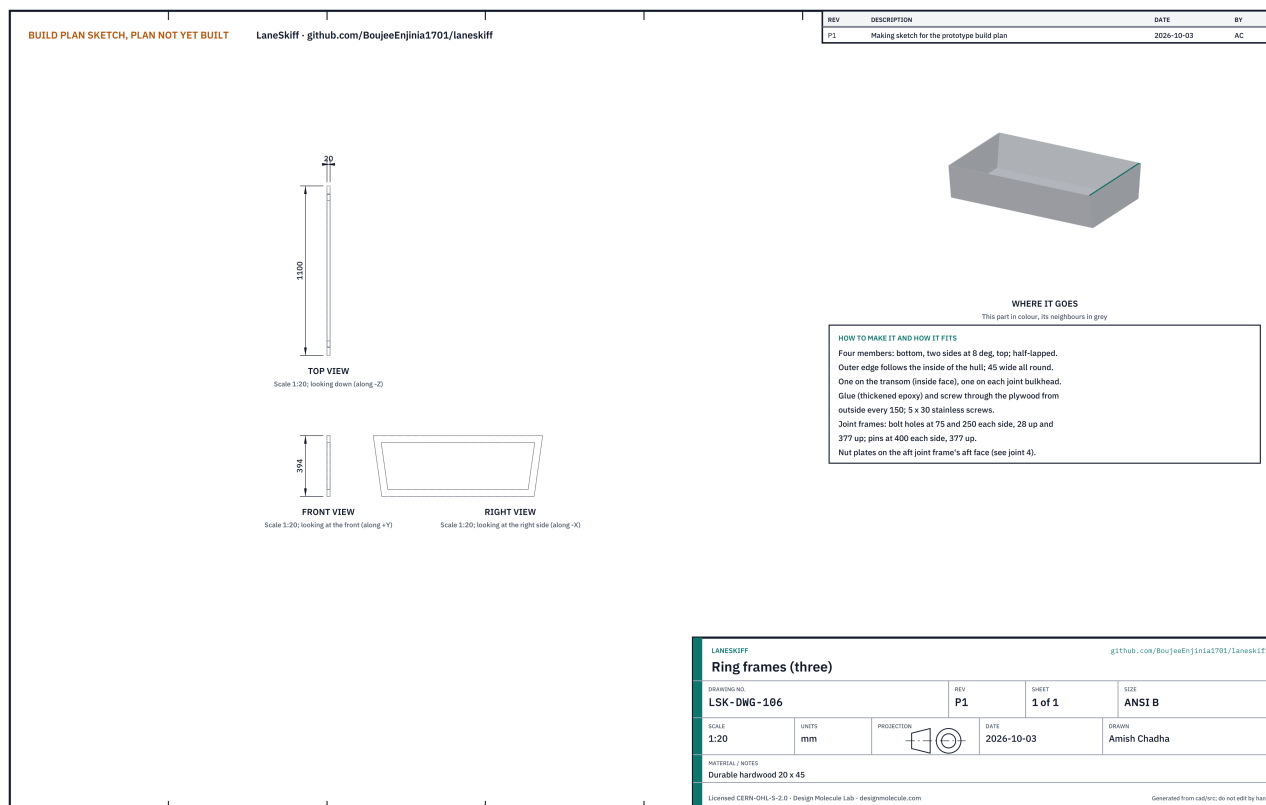
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/laneskiff

Figure 5. Aft ring frame, aft bulkhead, forward bulkhead, forward ring frame, clamped by an M10 bolt into a nut plate.

Check before moving on. With the two halves standing together, the bulkhead faces touch all round (no gap over 1 mm).

3.6 Ring frames



What it is and what it is made from. Three rings of durable hardwood 20 x 45 mm: one inside the stern transom and one on each joint bulkhead. Each ring is four members (bottom, two sides at 8 degrees, top), half-lapped at the corners, following the inside of the hull with a 45 mm wide face.

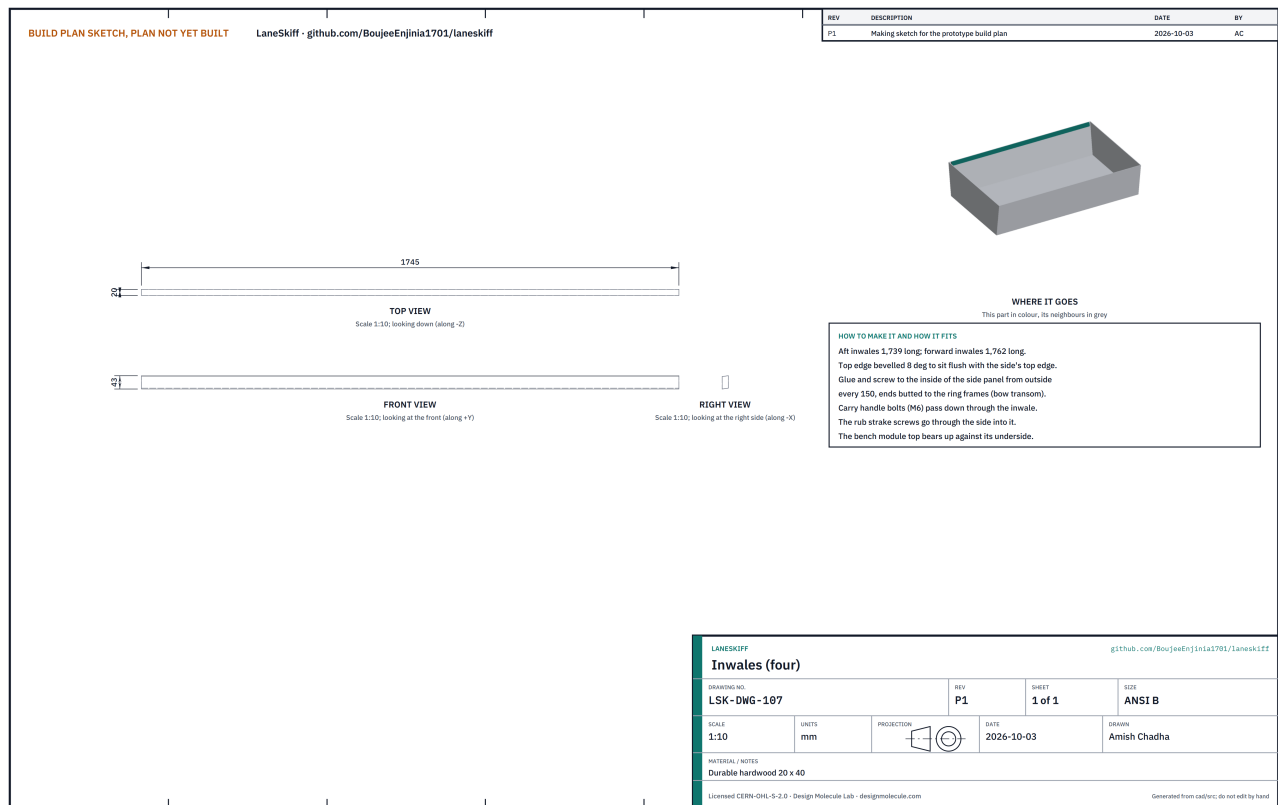
How to make it.

1. Cut the members, lay the ring on its bulkhead, mark and cut the half-laps.
2. Glue the ring to the plywood with thickened epoxy and screw through the plywood from outside every 150 mm with 5 x 30 stainless screws.
3. Joint frames only: with both halves clamped together, drill eight 11 mm bolt holes (at 75 mm and 250 mm each side of the centreline, 28 mm above the floor line and 377 mm up) and two 12.5 mm pin holes (400 mm each side, 377 mm up) through frame and bulkhead of both halves at once.
4. Screw a nut plate (a 40 x 40 x 3 mm steel plate with an M10 nut welded on) over each bolt hole on the aft face of the aft joint frame. Epoxy the two pins into the aft frame.

How it fits the parts next to it. The frames butt against the side panels, bottom and transom or bulkhead; the inwales and stringers end against their faces.

Check before moving on. A bolt runs freely through all four layers into every nut plate by hand.

3.7 Inwales



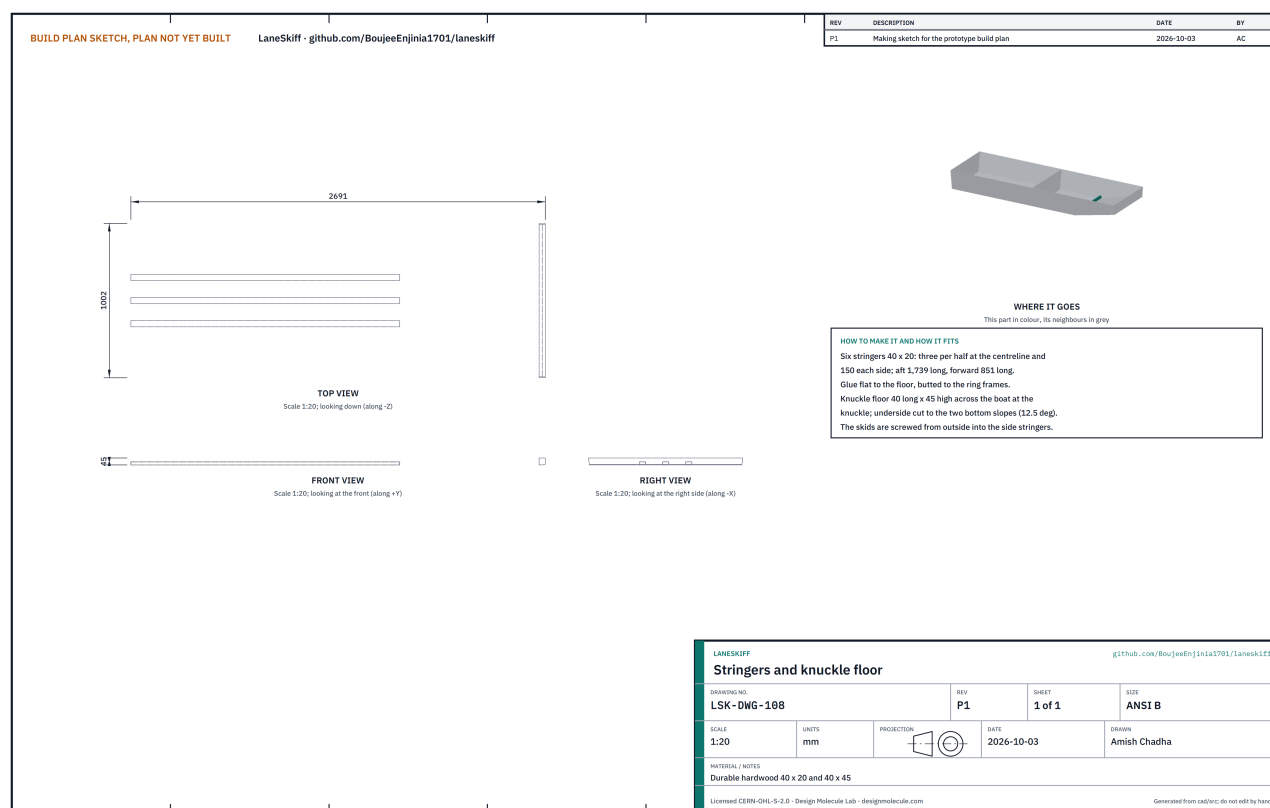
What it is and what it is made from. Four lengths of durable hardwood 20 x 40 mm: aft inwales 1,739 mm, forward inwales 1,762 mm.

How to make it. Bevel the top edge 8 degrees; glue to the inside of the side panel along the inwale line, screwing from outside every 150 mm; ends butt the ring frames (and the bow transom forward).

How it fits the parts next to it. Its top is flush with the side's top edge. The carry handle bolts pass down through it, the rub strake screws go into it, and the top of each bench module bears up against its underside.

Check before moving on. Top edge flush with the side within 1 mm along its length.

3.8 Stringers and knuckle floor



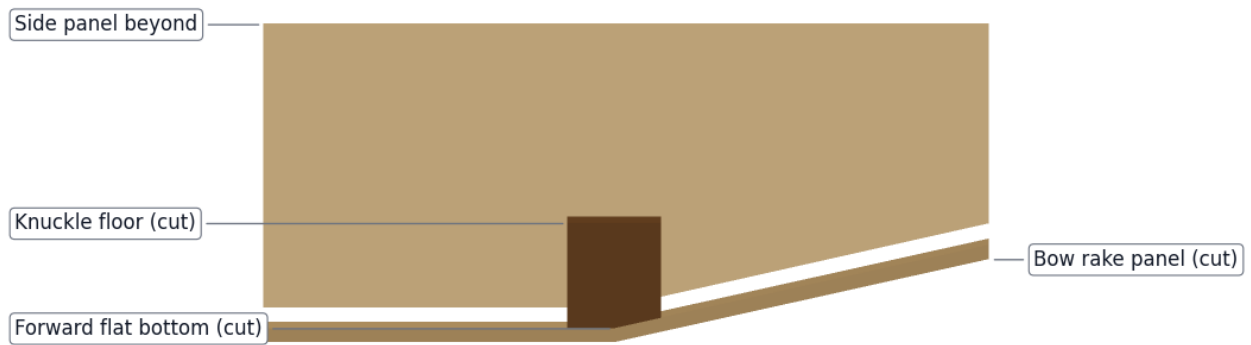
What it is and what it is made from. Six hardwood stringers 40 x 20 mm (three per half: on the centreline and 150 mm each side; 1,739 mm long aft, 851 mm forward) and a knuckle floor 40 mm long and 45 mm high across the forward half at the knuckle, its underside cut to the two bottom slopes.

How to make it. Glue the stringers flat to the floor on their marked lines, ends butted to the ring frames and the knuckle floor. Glue and screw the knuckle floor across both bottom panels at the knuckle.

How it fits the parts next to it. The bow rake panel and the forward flat bottom butt at the knuckle on top of the knuckle floor; tape outside and fillets inside close the seam.

Joint 3: bottom knuckle, cut along the boat

Butted panels on the knuckle floor; tape outside, fillet inside



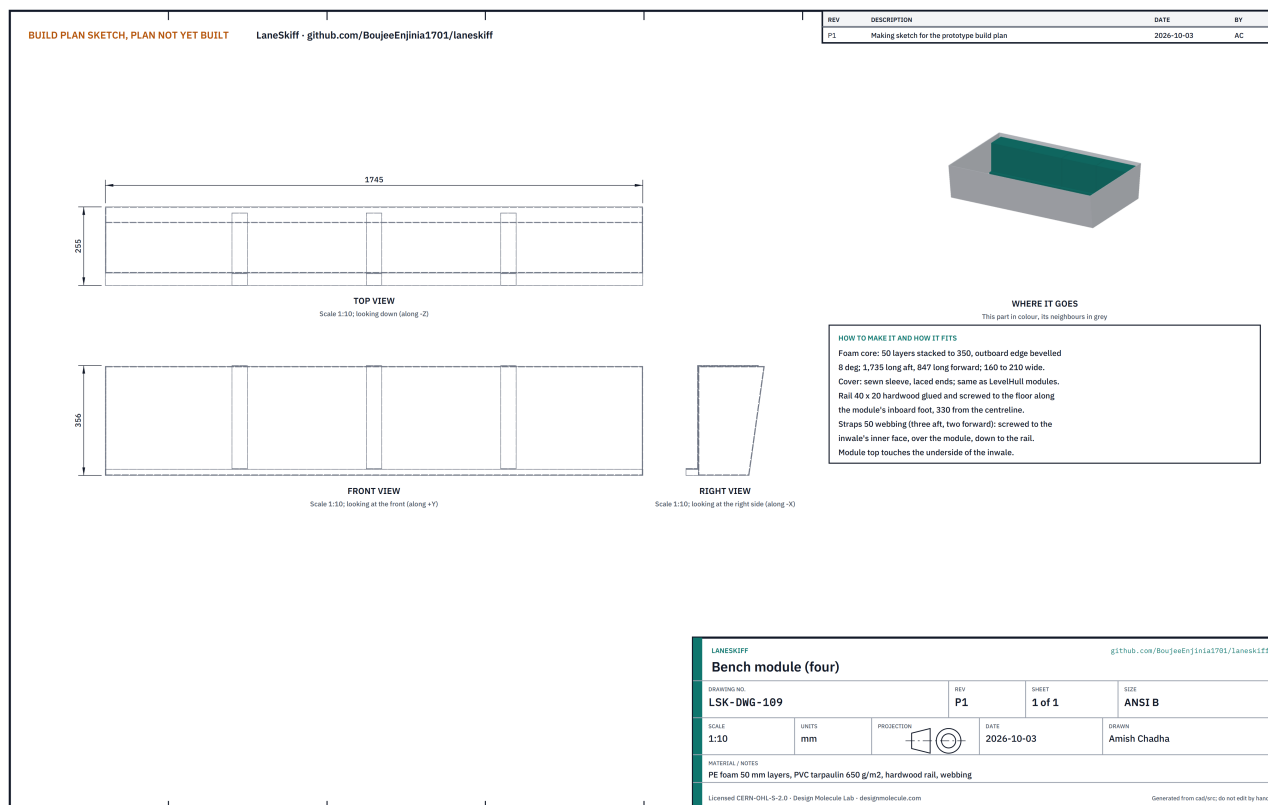
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/laneskiff

Figure 6. The knuckle: the two bottom panels butt on the knuckle floor.

Check before moving on. No stringer rocks; fillets continuous.

3.9 Bench modules



What it is and what it is made from. Four modules, one each side of each half: a core of closed-cell polyethylene foam in 50 mm layers, stacked 350 mm high, with its outboard edge bevelled 8 degrees to lie on the side (160 to 210 mm wide, 1,735 mm long aft, 847 mm forward); a sewn sleeve of PVC-coated polyester tarpaulin with laced ends, as for the LevelHull modules; a hardwood bench rail 40 x 20 mm on the floor along the module's inboard foot, 330 mm from the centreline; and webbing straps (three per aft module, two per forward module).

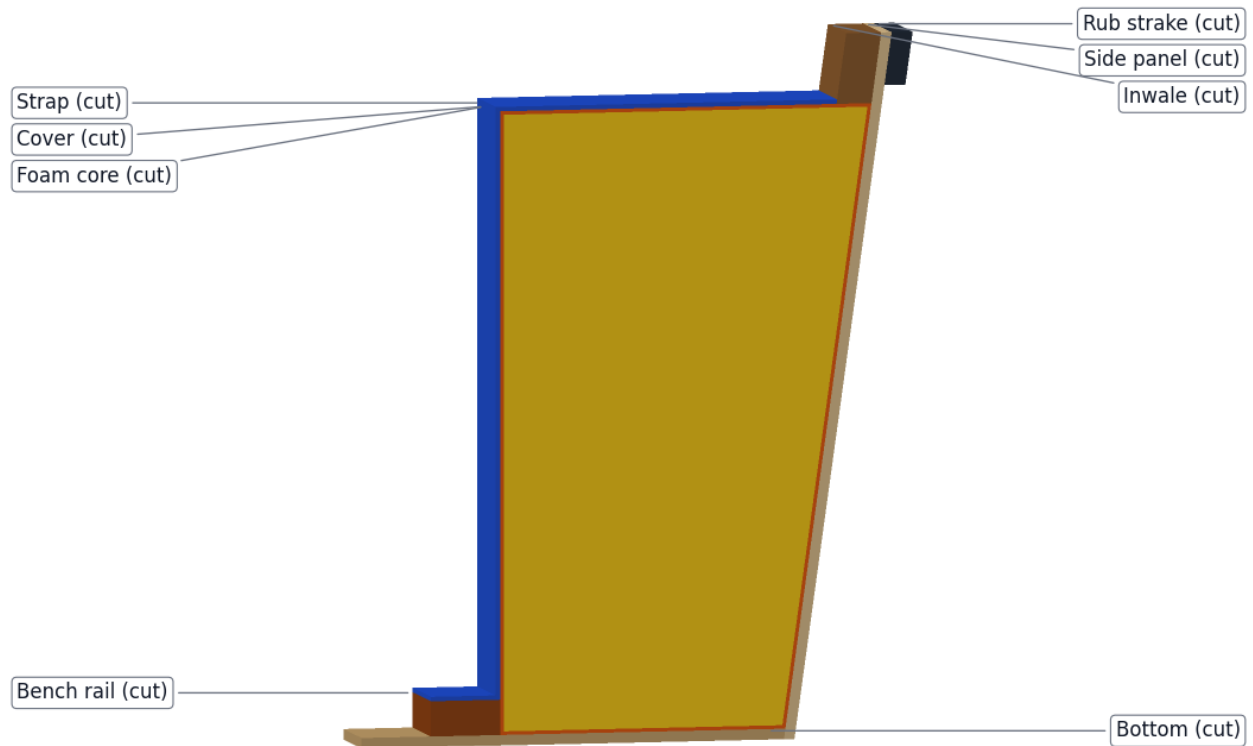
How to make it.

1. Cut the foam layers, bevel the outboard edges, stack them; no glue.
2. Sew the sleeve to the stack size with two rows of UV-stabilised stitching; slide the stack in and lace the ends.
3. Glue and screw the bench rails to the floor.
4. Cut the straps 50 mm webbing, heat-seal the ends.

How it fits the parts next to it. The module sits on the floor against the side panel, between the ring frames, with its top touching the underside of the inwale. Each strap is screwed to the inwale's inner face with two stainless screws and a penny washer, runs over the module top and down its inboard face, and is screwed to the top of the bench rail.

Joint 5: bench module, cut across at a strap

Module tight under the inwale; strap screwed to the inwale and the rail



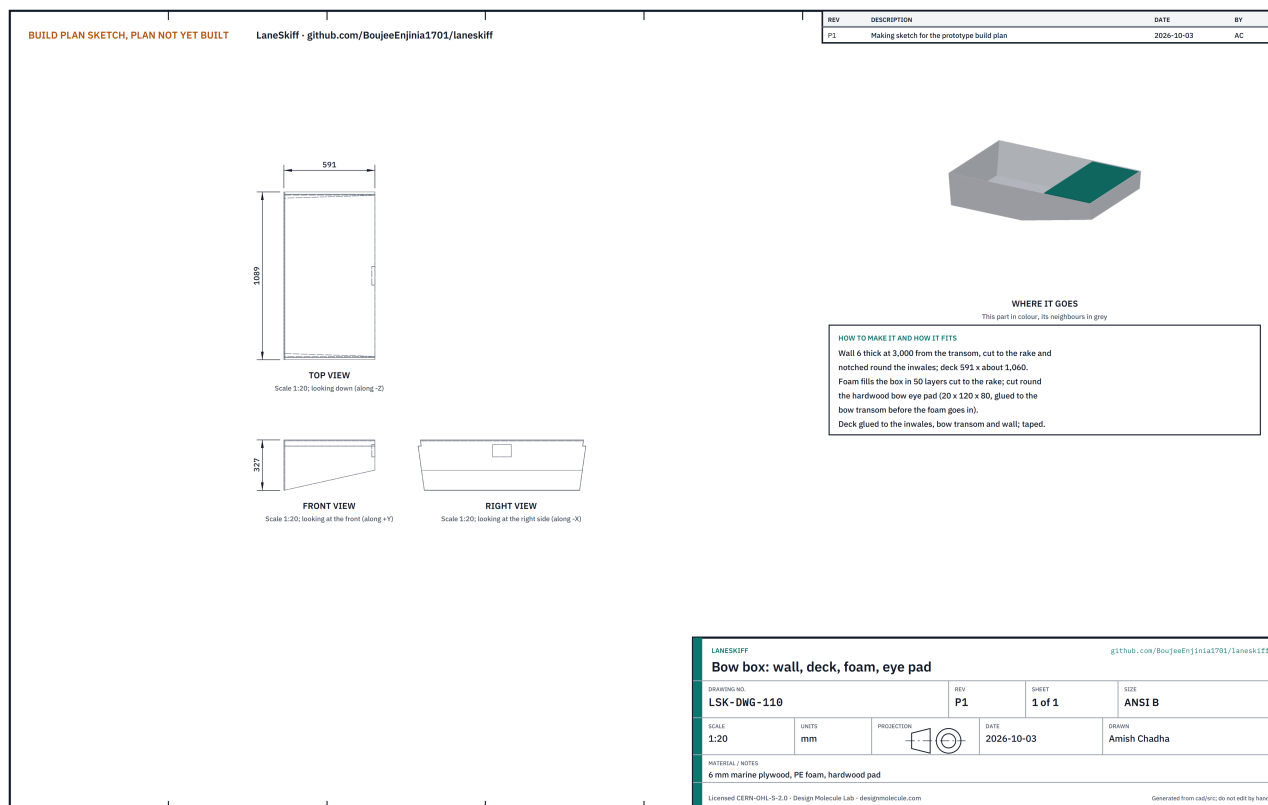
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/laneskiff

Figure 7. Bench module tight under the inwale, strapped to the inwale and the bench rail.

Check before moving on. The module cannot be lifted by hand at any strap.

3.10 Bow box



What it is and what it is made from. A 6 mm plywood wall standing 3.0 m forward of the transom, cut to the rake and notched round the inwales; a 6 mm deck 591 mm long and about 1,060 mm wide; foam layers filling the box; and a hardwood pad 20 x 120 x 80 mm for the bow eye.

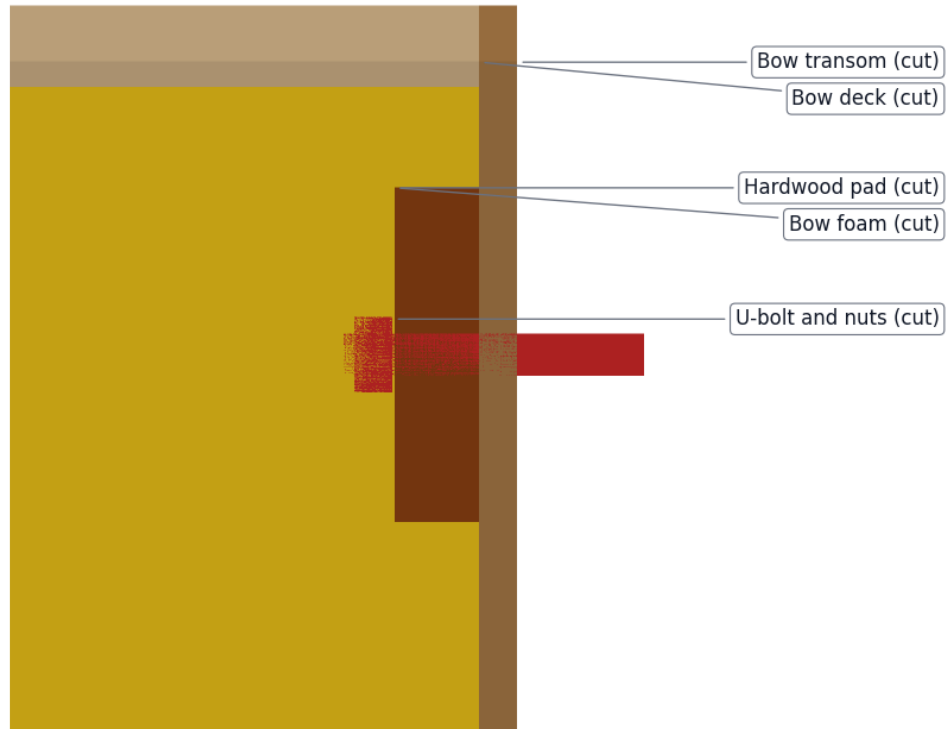
How to make it.

1. Glue the pad to the inside of the bow transom; fit the bow eye (a stainless U-bolt) through transom and pad with its backing plate and nuts inside.
2. Glue in the wall; fillet and tape it to the bottom and sides.
3. Cut the foam layers to the rake and round the pad and nuts; pack the box.
4. Glue the deck to the inwales, bow transom and wall; tape the seams.

How it fits the parts next to it. The deck lies between the inwales, flush with their tops; the foam fills the box with no voids.

Joint 8: bow eye, cut through one leg

U-bolt through the bow transom and pad; nuts inside, fitted before the foam



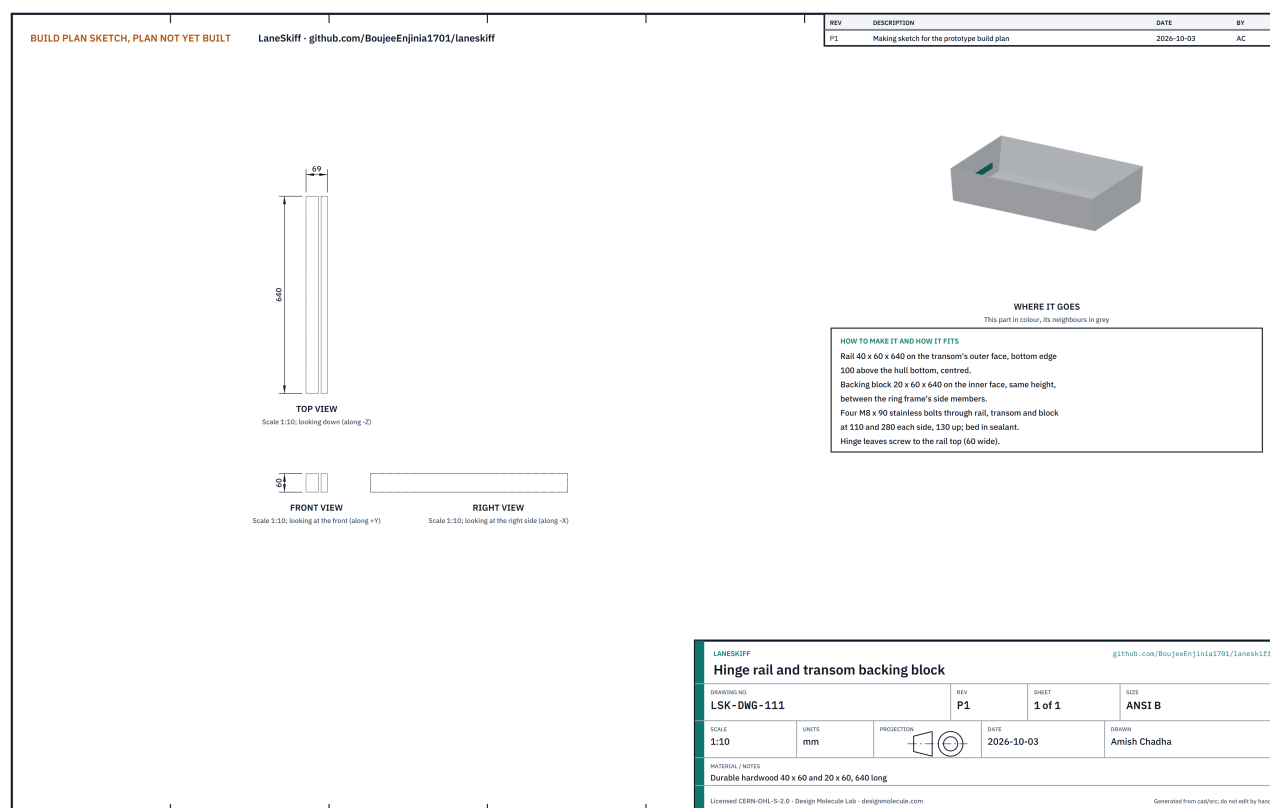
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/laneskiff

Figure 8. Bow eye through the bow transom and pad, nuts inside, foam round it.

Check before moving on. Bow eye nuts tight before the foam goes in; the eye does not move under a hard pull by hand.

3.11 Hinge rail and transom backing block



What it is and what it is made from. A hardwood rail 40 x 60 x 640 mm on the transom's outer face, its bottom edge 100 mm above the bottom of the hull, centred; and a hardwood backing block 20 x 60 x 640 mm on the inner face at the same height, between the ring frame's side members.

How to make it. Cut, seal, glue the backing block inside; drill through block, transom and rail together for four M8 x 90 stainless bolts at 110 mm and 280 mm each side, 130 mm up; bed the bolts in sealant.

How it fits the parts next to it. The strap hinges screw to the rail's top face (60 mm wide).

Joint 6: hinge rail and step hinge, cut along the boat

40 mm gap between rail and float; hinge pin over the gap; M8 bolts through rail, transom, block



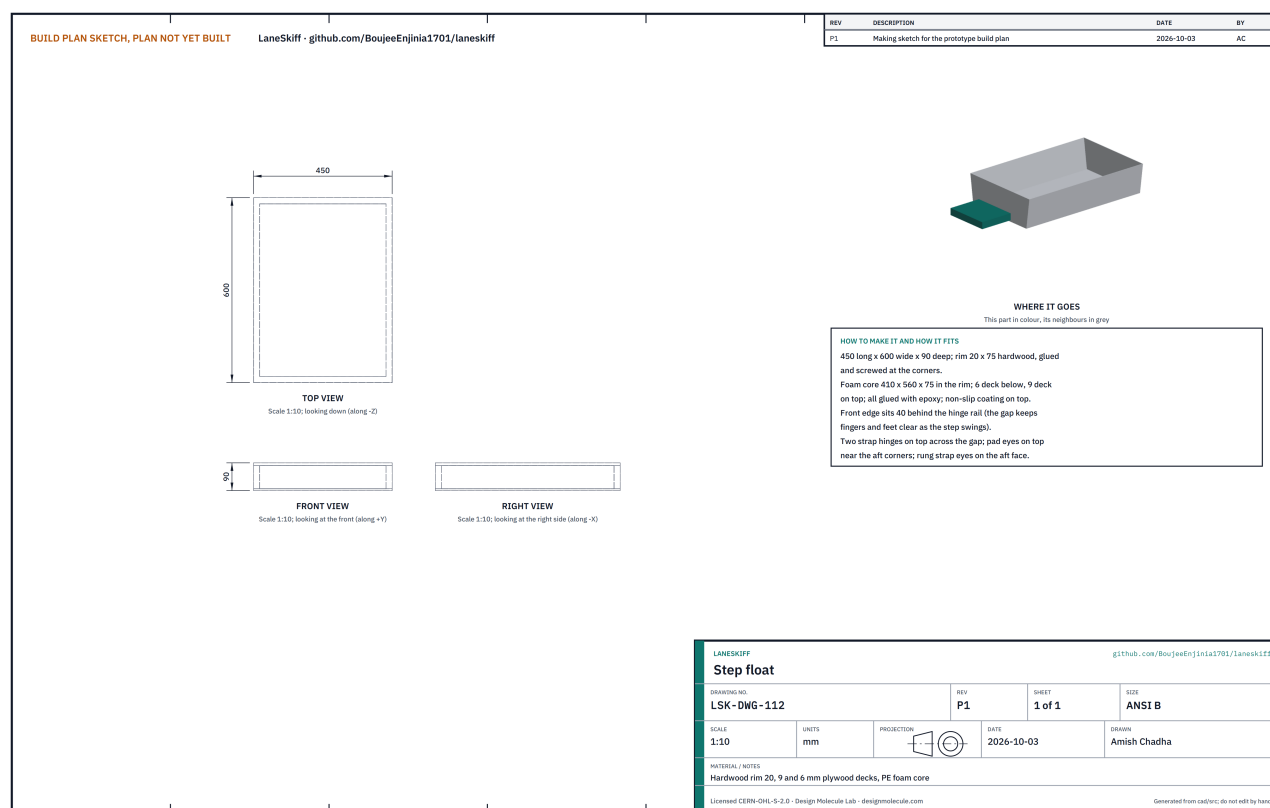
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/laneskiff

Figure 9. Hinge rail bolted through transom and backing block; the step hinge crosses the 40 mm gap to the float.

Check before moving on. The rail does not move when levered with a 1 m bar.

3.12 Step float



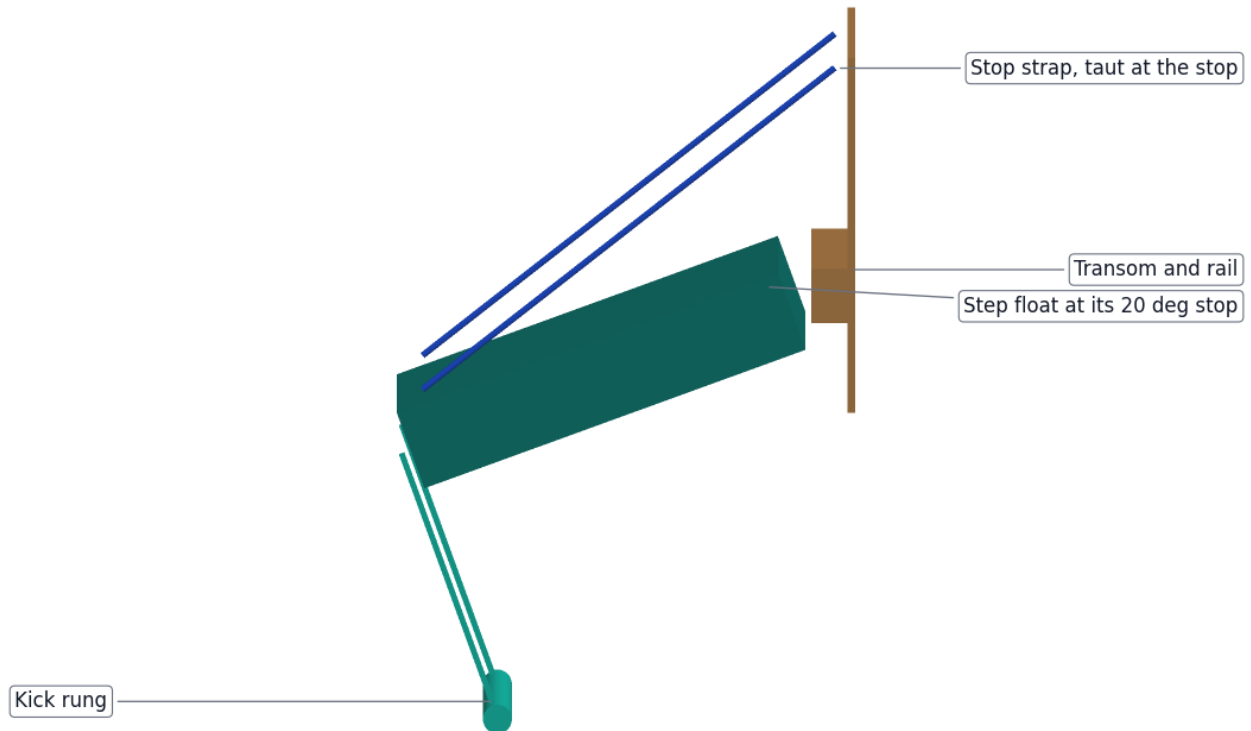
What it is and what it is made from. A float 450 mm long, 600 mm wide and 90 mm deep; a hardwood rim 20 x 75 mm, a foam core 410 x 560 x 75 mm, a 6 mm plywood bottom deck and a 9 mm plywood top deck, all glued with epoxy; non-slip coating on top.

How to make it. Glue and screw the rim at the corners; glue the bottom deck on; set the foam in; glue the top deck on; coat. Fit two pad eyes on top near the aft corners and two strap eyes on the aft face.

How it fits the parts next to it. Its front edge sits 40 mm behind the hinge rail. Two heavy stainless strap hinges, one leaf on the rail top and one on the float top, cross the gap with their pins over it. Two webbing stop straps run from pad eyes high on the transom to the pad eyes on the float and hold it at 20 degrees below level at most.

Joint 7: the step at its stop, from the side

Stop straps from the transom pad eyes hold the float 20 deg below level



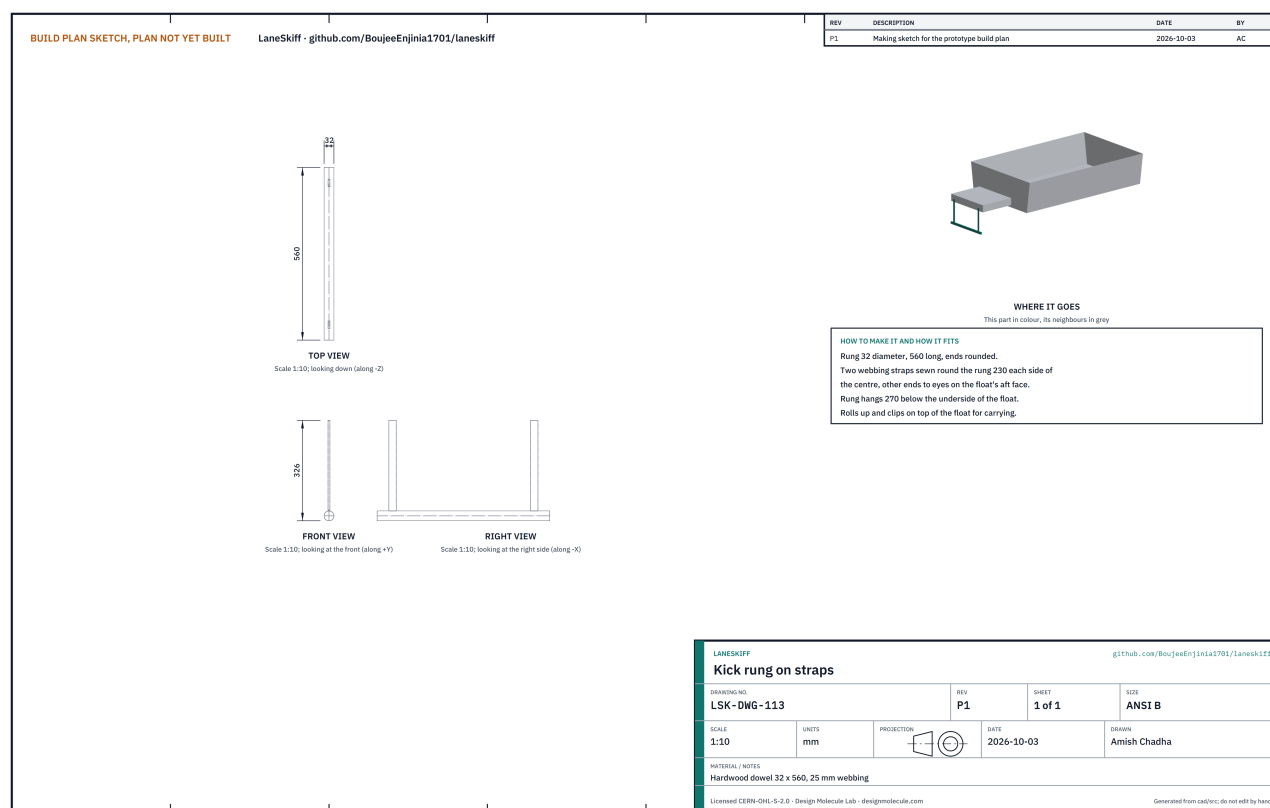
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/laneskiff

Figure 10. The step swung down to its stop, with the kick rung below.

Check before moving on. The float swings from folded up to its stop without touching the rail or transom; nothing closes to less than 25 mm.

3.13 Kick rung



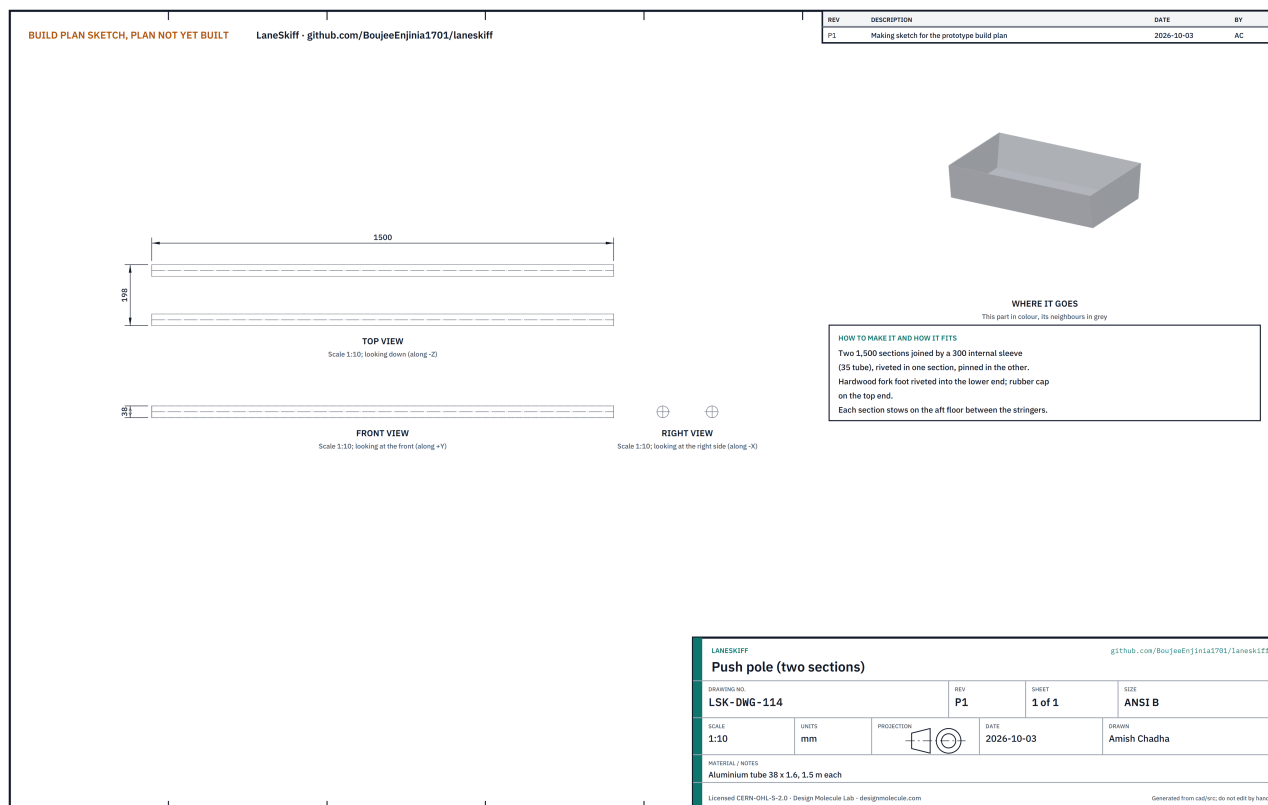
What it is and what it is made from. A hardwood rung 32 mm in diameter and 560 mm long on two 25 mm webbing straps.

How to make it. Round the ends; sew each strap round the rung 230 mm each side of the centre; sew the other ends to the strap eyes on the float's aft face so the rung hangs 270 mm below the float's underside.

How it fits the parts next to it. It rolls up and clips on top of the float for carrying.

Check before moving on. Each strap holds a person standing on the rung (see section 6).

3.14 Push pole



What it is and what it is made from. Two 1,500 mm sections of aluminium tube 38 x 1.6 mm, joined by a 300 mm internal sleeve riveted into one section and pinned into the other; a hardwood fork foot riveted into the lower end; a rubber cap on the top end.

How to make it. Cut, deburr, rivet the sleeve, drill the cross pin, rivet the foot.

How it fits the parts next to it. Each section stows on the aft floor between the stringers.

Check before moving on. The joined pole does not wobble at the joint.

3.15 Bought components

- **Foam sheet:** closed-cell polyethylene, 2,000 x 1,000 x 50 mm, about 30 kg/m³; not polystyrene (fuel dissolves it) and not open-cell foam.
- **HDPE strip:** rub strakes 12 x 30 mm screwed through the side into the inwale every 200 mm; skids 40 x 8 mm screwed into the side stringers every 150 mm with 5 x 30 stainless screws, each hole bedded in sealant; the forward skids bend over the knuckle.
- **Joint bolts and nut plates:** M10 x 80 stainless A4-70 hex bolts with 30 mm washers; nut plates welded by a local welder and galvanised.
- **Hinges and pad eyes:** two heavy stainless strap hinges about 100 mm wide with 6 mm pins; four stainless pad eyes.
- **Webbing:** 25 mm polyester for the stop and rung straps, 50 mm for the bench straps and slings.

- **Handles:** two stainless tube grab handles about 250 mm long for the transom; eight grab handles about 200 mm long for the inwales, each with two M6 bolts.
- **Paddles, bow line and slings:** two 1.4 m paddles, 10 m of 10 mm polyester rope, two padded slings with snap hooks.

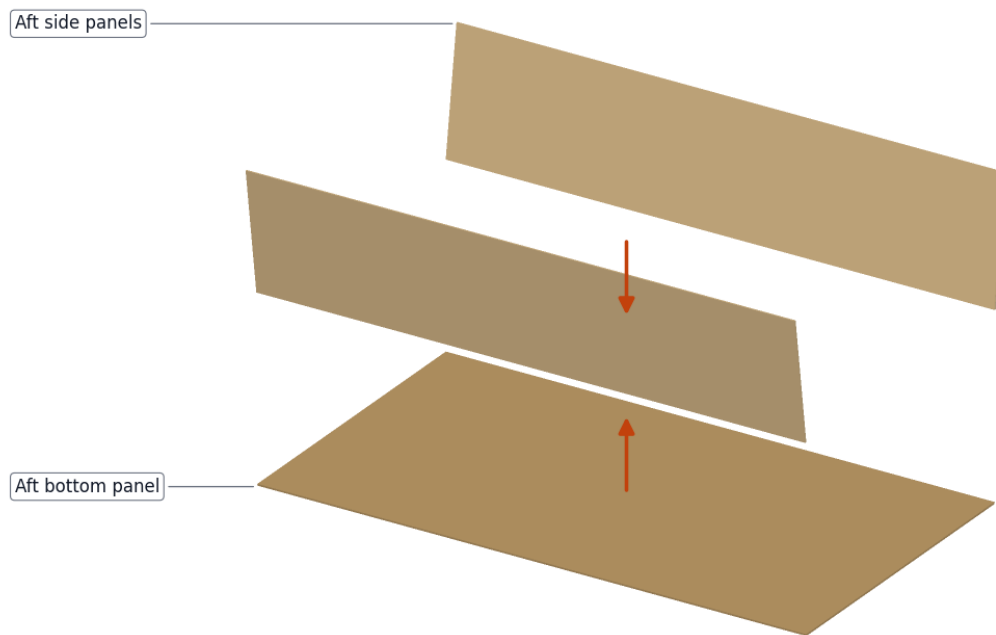
■ 4. Putting it together

Each picture shows the parts already fitted in grey and the parts being fitted in colour, with an arrow showing the way they go in.

Step 1: stitch the aft bottom and sides

Step 1: stitch the aft bottom and sides

Sides outside the bottom edges, stitched every 150 mm; check the diagonals are equal



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

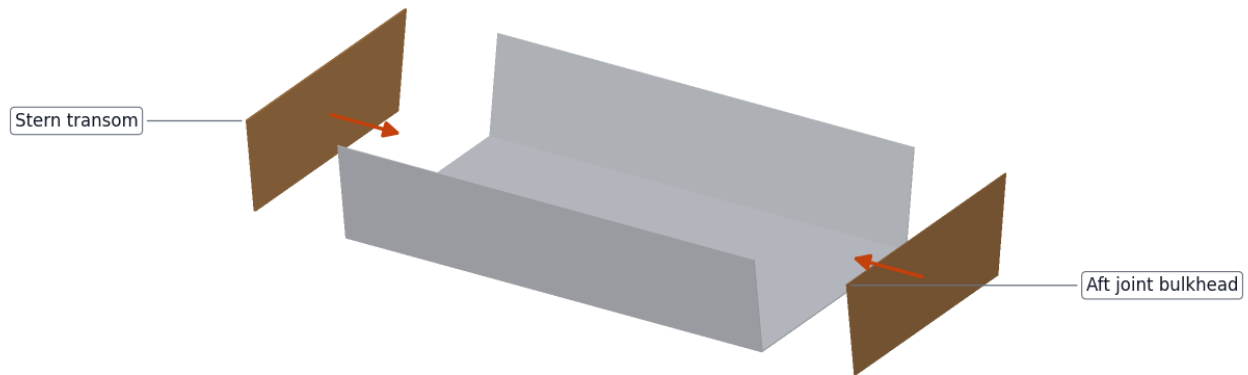
github.com/BoujeeEnjinia1701/laneskiff

Stitch the side panels to the bottom with copper wire or cable ties every 150 mm. Check the diagonals are equal before any glue.

Step 2: fit the stern transom and the aft joint bulkhead

Step 2: fit the stern transom and the aft joint bulkhead

Both sit on the bottom between the sides; tack with fillets, then tape inside



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

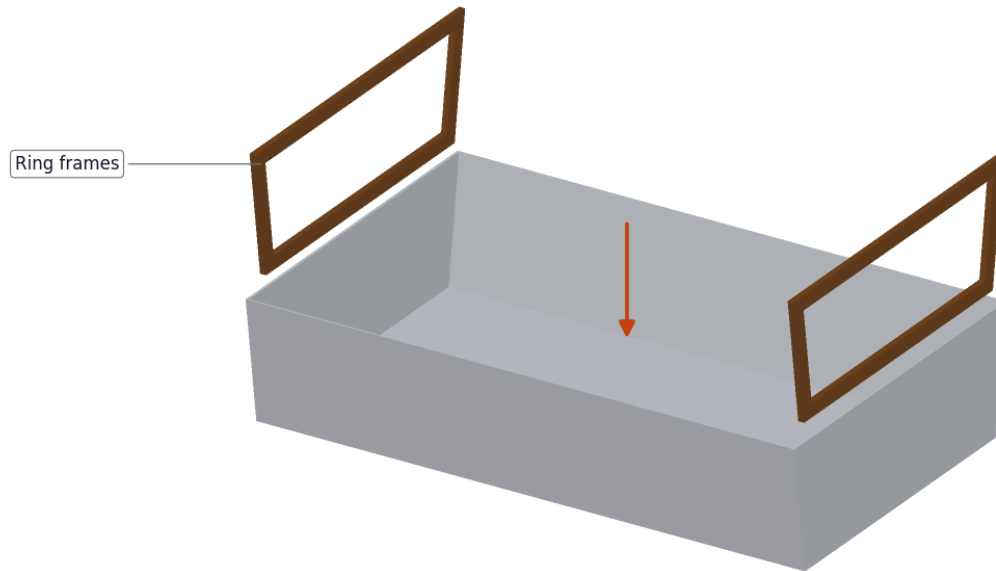
github.com/BoujeeEnjinia1701/laneskiff

Both sit on the bottom between the sides. Tack with small fillets, check square, then fillet and tape every inside seam.

Step 3: glue in the ring frames

Step 3: glue in the ring frames

Transom frame and joint frame, glued and screwed through the plywood



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

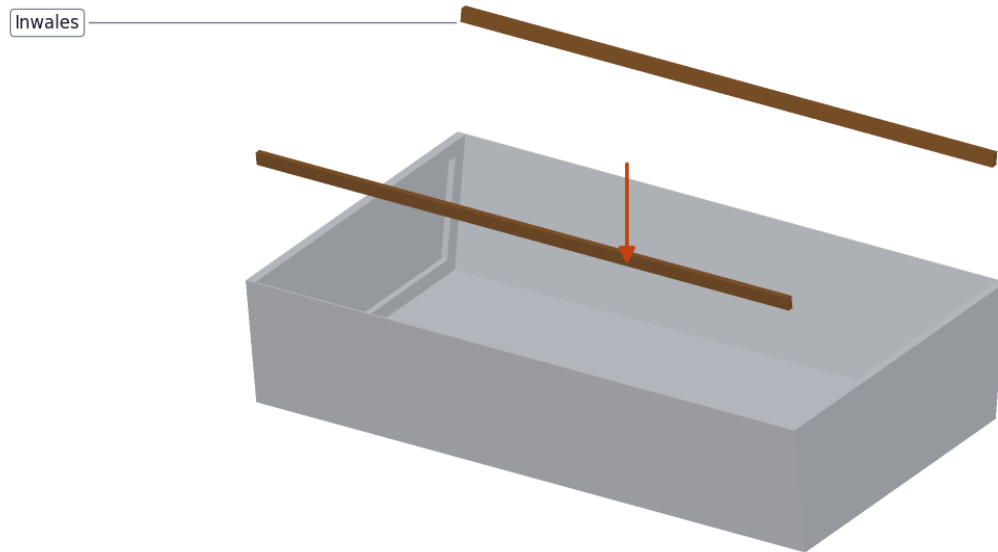
github.com/BoujeeEnjinia1701/laneskiff

Glue and screw the transom frame and the aft joint frame. Hold point: let the epoxy cure before drilling.

Step 4: fit the inwales

Step 4: fit the inwales

Inside the top of each side, frame to frame; top edges flush



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

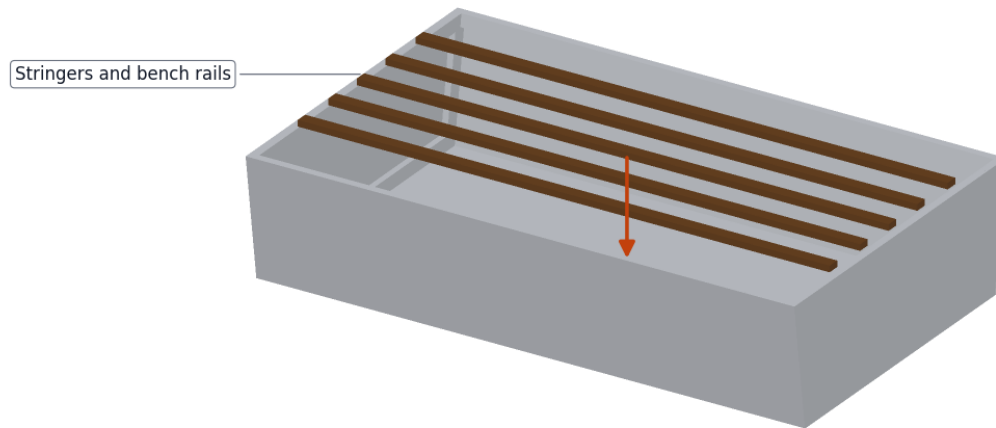
github.com/BoujeeEnjinia1701/laneskiff

Glue and screw inside the top of each side, frame to frame.

Step 5: glue down the stringers and bench rails

Step 5: glue down the stringers and bench rails

Three stringers and two bench rails on the floor, butted to the frames



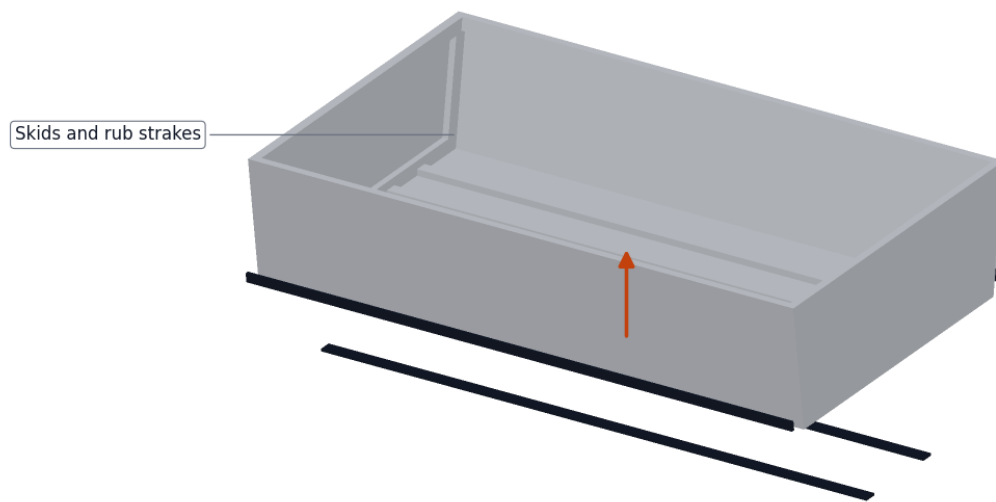
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/laneskiff

Three stringers and two bench rails on their marked lines.

Step 6: glass the outside, then fit the skids and strakes**Step 6: glass the outside, then fit skids and strakes**

Turn the half over: glass the bottom and sides, then screw the skids into the stringers



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

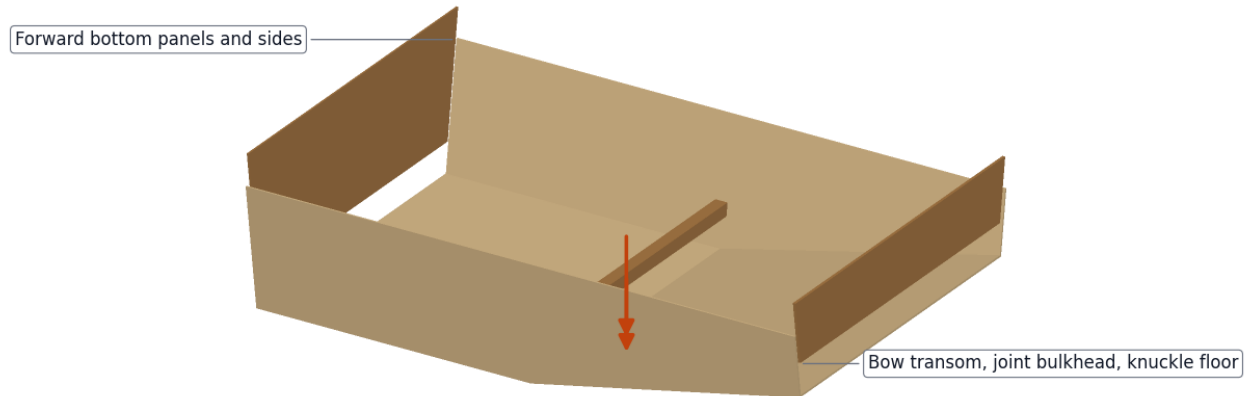
github.com/BoujeeEnjinia1701/laneskiff

Turn the half over, remove the stitches, fill and round the chines, and glass the bottom and sides with 400 g/m² cloth. When cured, screw the skids into the side stringers and the rub strakes into the inwales, each hole bedded in sealant. Paint.

Step 7: build the forward half the same way

Step 7: build the forward half the same way

Bottom panels butted at the knuckle on the knuckle floor; bow transom on the rake



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

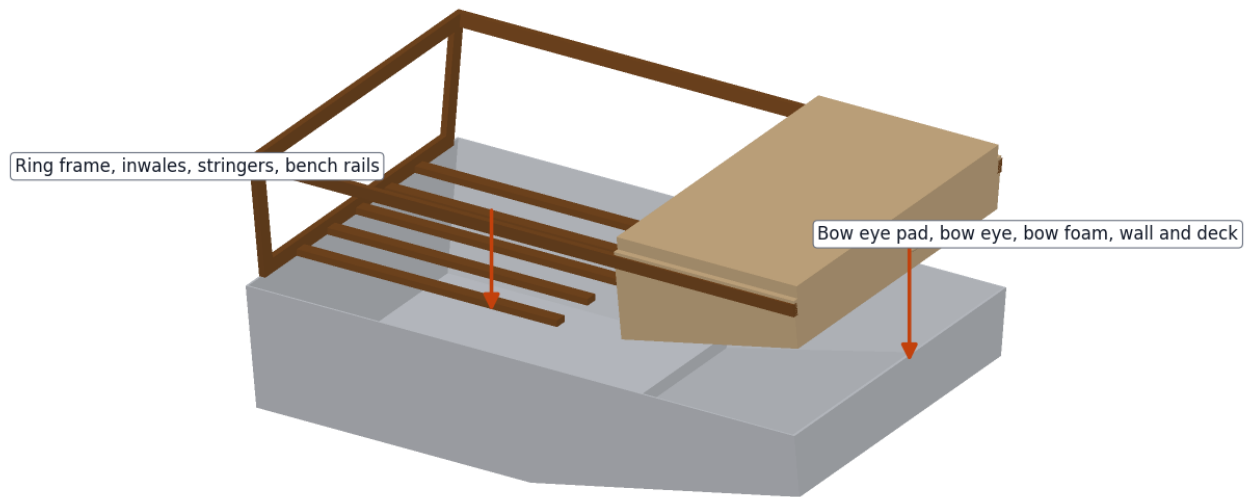
github.com/BoujeeEnjinia1701/laneskiff

As steps 1 and 2, with the two bottom panels butted at the knuckle and the bow transom on the rake.

Step 8: frame the forward half and close the bow box

Step 8: frame the forward half and close the bow box

Bow eye and pad before the foam; foam layers, then the deck glued on



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

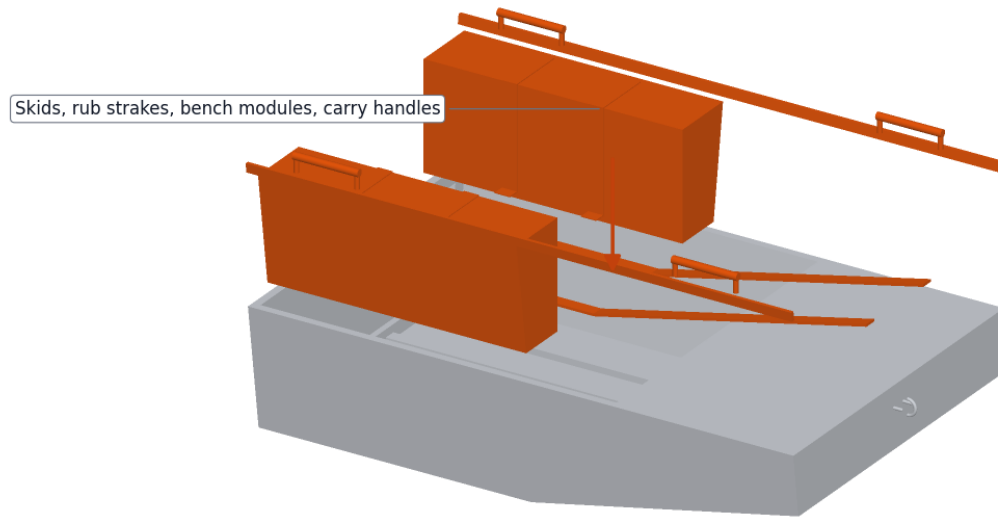
github.com/BoujeeEnjinia1701/laneskiff

Ring frame, inwales, stringers and bench rails as on the aft half; then the bow eye and pad, the bow box wall, the foam and the deck.

Step 9: glass, skids, strakes, bench modules and handles

Step 9: glass, skids, strakes, bench modules and handles

As steps 6 and 10 to 11 on the aft half



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

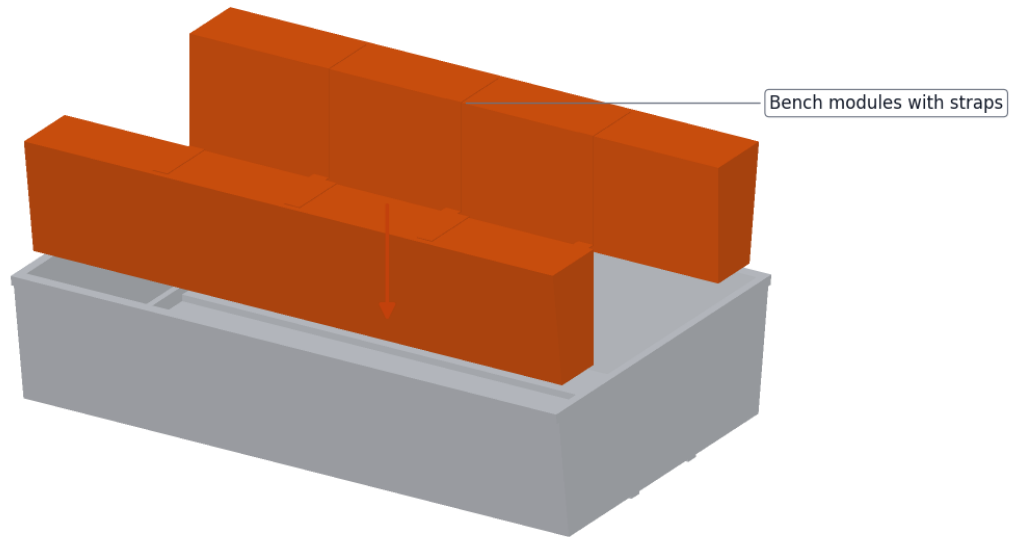
github.com/BoujeeEnjinia1701/laneskiff

Finish the forward half as in steps 6, 10 and 11.

Step 10: strap in the aft bench modules

Step 10: strap in the aft bench modules

Module tight under the inwale; straps screwed to inwale and rail



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

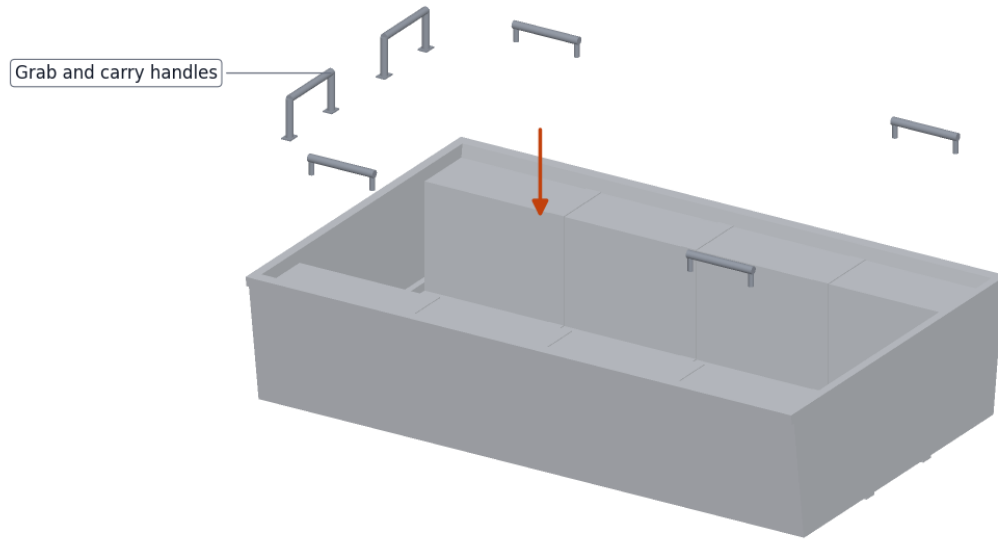
github.com/BoujeeEnjinia1701/laneskiff

Slide each module under the inwale, against the side; screw each strap to the inwale and the bench rail, pulled snug.

Step 11: fit the grab and carry handles

Step 11: fit the grab and carry handles

M6 bolts down through the inwales; transom handles through the frame



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

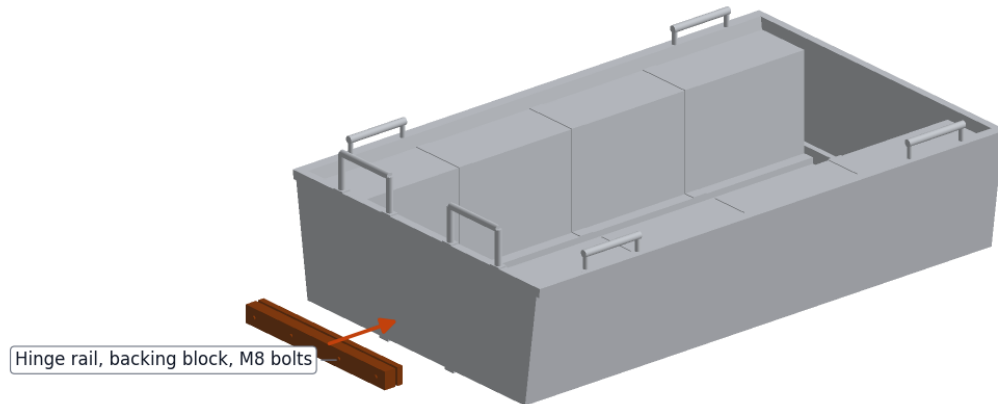
github.com/BoujeeEnjinia1701/laneskiff

Carry handles with M6 bolts down through the inwales, two at each end of each half; transom grab handles through the transom and its ring frame.

Step 12: bolt on the hinge rail

Step 12: bolt on the hinge rail

Four M8 bolts through rail, transom and backing block, bedded in sealant



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

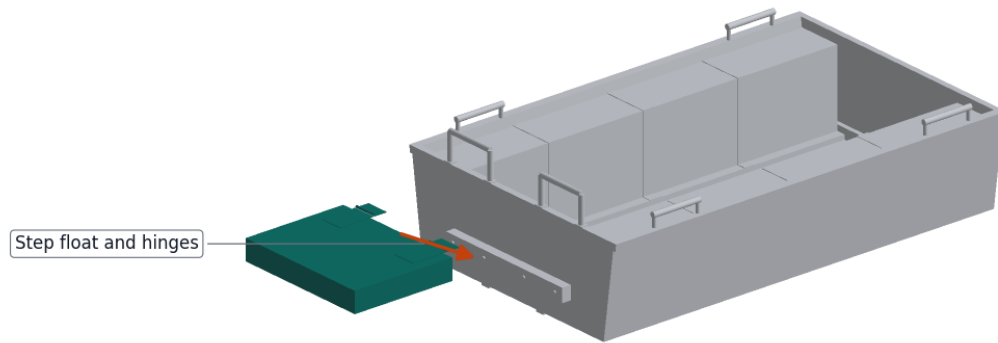
github.com/BoujeeEnjinia1701/laneskiff

Four M8 bolts through rail, transom and backing block, bedded in sealant.

Step 13: hang the step float

Step 13: hang the step float

Hinge leaves on the rail top and the float top; 40 mm gap



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

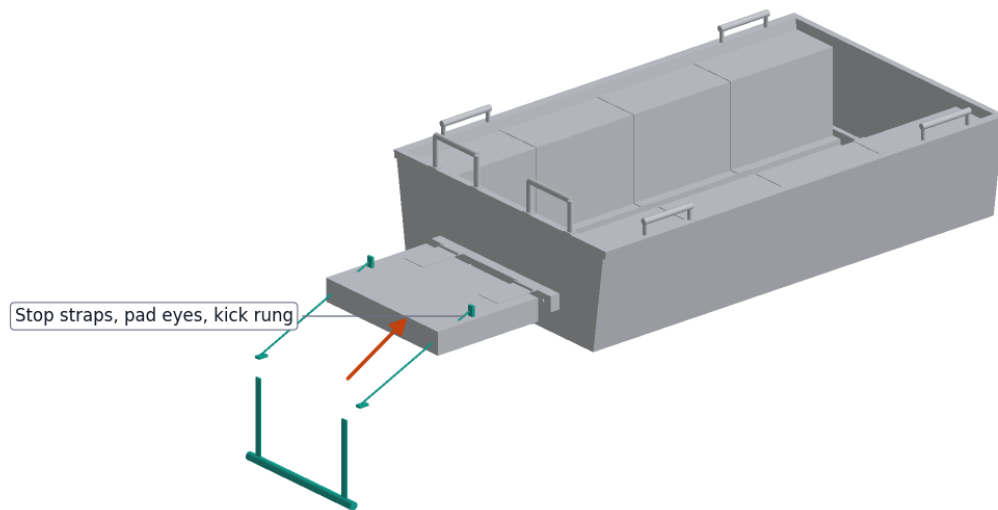
github.com/BoujeeEnjinia1701/laneskiff

Screw the hinge leaves to the rail top and the float top with the 40 mm gap set by two spacer blocks; remove the blocks.

Step 14: fit the stop straps and the kick rung

Step 14: fit the stop straps and the kick rung

Float stops 20 deg below level; rung 270 mm below it



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

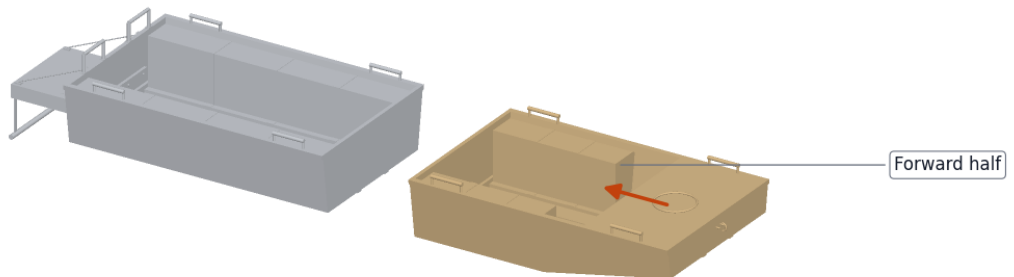
github.com/BoujeeEnjinia1701/laneskiff

Pad eyes on the transom; adjust the straps so the float stops 20 degrees below level; hang the rung 270 mm below the float.

Step 15: bring the halves together

Step 15: bring the halves together

Pins into their holes, bulkhead faces touching



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

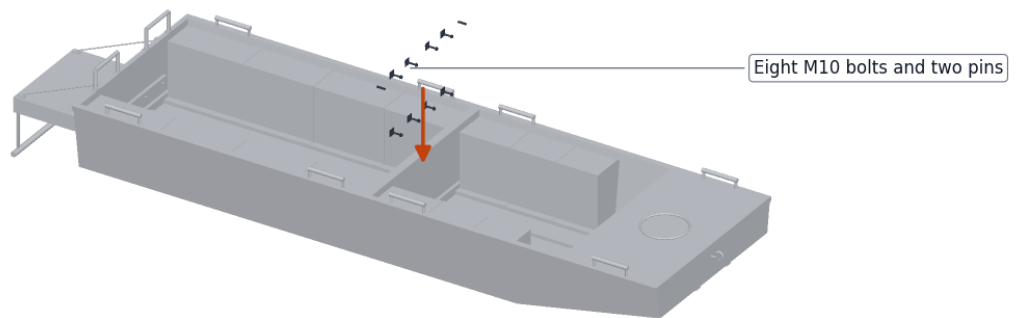
github.com/BoujeeEnjinia1701/laneskiff

At the water's edge, stand the halves end to end, slide the pins into their holes and close the bulkhead faces together.

Step 16: bolt the halves together

Step 16: bolt the halves together

Bottom row first, then top; snug with one 17 mm spanner



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

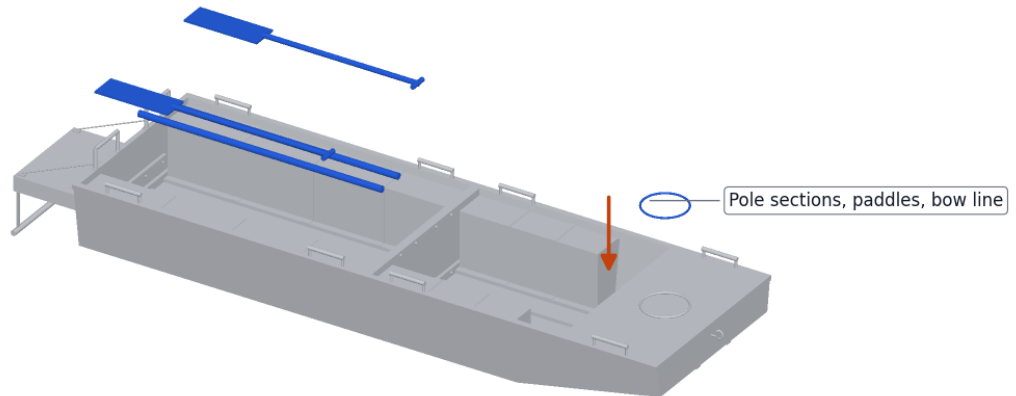
github.com/BoujeeEnjinia1701/laneskiff

Bottom row first, then the top row, snugged with one 17 mm spanner from the forward side. Hold point: all eight bolts in before anyone steps in.

Step 17: stow the pole, paddles and bow line

Step 17: stow the pole, paddles and bow line

Pole sections on the aft floor, paddles on the aft benches, line on the bow deck



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/laneskiff

Pole sections on the aft floor between the stringers, paddles on the aft benches, the bow line coiled on the bow deck and tied to the bow eye.

■ 5. First checks

These are listed here and recorded in a TRL 4 test report.

Table 2. First checks.

Check	Requirement	How	Pass when
Beam	R4	Tape over the rub strakes; carry through a 1.3 m gate	Fits the gate without scraping
Carry mass	R1	Weigh each half on a hanging scale	Recorded against LSK-CAL-001 (56 and 45 kg expected)
Joining the halves	R7	Two people, one spanner, timed from halves on the ground to eight bolts tight	Joined, and within the R7 time
Each half alone	Safety	Float each half on its own in calm, shallow water	Floats upright, no leak at the bulkhead
Load and draft	R2, R3	Water ballast in containers placed as the rated persons, in calm water	Drafts recorded at the transom and bow; no water over the gunwale
Swamp	R5	Fill with water, ballast for the rated persons at two thirds, following the ABYC H-8 procedure in a pool	Floats, trim and freeboard recorded at the transom and bow
Boarding	R6	Several volunteers of different builds board over the stern step from waist-deep water, inclinometer on the hull	Heel recorded; every volunteer boards unaided
Step proof load	R6, safety	Twice the working load on the float aft edge at its stop (CalRig)	No damage, no hinge or strap movement

■ 6. Safety stops

Work stops at each point below until what is listed is true.

- **Before anyone stands in the boat:** all eight joint bolts in and tight; every hull seam inspected, no dry fillets.
- **Before the first time on the water:** each half floated alone without a leak; the bow eye pulled to twice the towing load.
- **Before any boarding trial:** the step hinges, stop straps and kick rung proof-loaded to twice the working load (an 80 kg person kneeling on the float's aft edge with a factor of two); the hinge gap checked at 40 mm with nothing closing below 25 mm through the swing.
- **Before any load or swamp trial:** calm, shallow water no deeper than chest height, no current; a safety boat or bank team with a throw line; every person in a life jacket and briefed; ballast in place of people for the first swamp test.
- **Before any rescue use:** the TRL 4 trials passed, the rating decided by Amish and marked on the transom, and the crew trained under its own agency's rules. Never in moving water.

■ 7. Tools, skills and workspace

- A covered, ventilated space about 5 x 3 m where epoxy can cure above 15 degrees Celsius, with two trestles.
- Jigsaw or circular saw, block plane, drill and drivers, a sanding block or orbital sander, clamps (twelve or more), mixing pots and fillet spatulas, a tape and long straightedge, a bevel gauge, a 17 mm spanner and a 13 mm spanner.
- Skills: stitch-and-glue boatbuilding (any small boatyard), basic sewing with a heavy machine or a sail maker for the covers and straps, and a welder for the nut plates.

- Gloves, eye protection and a dust mask for epoxy, glass and sanding.

■ 8. Where the numbers come from

- Model: `cad/src/model.py` (dimensions, constructability checks), `cad/step/` and `cad/stl/` .
- Drawings: `cad/drawings/LSK-DWG-001` (general arrangement, Rev P2) and `LSK-DWG-101` to `114` (making sketches).
- Calculations: `docs/04-calcs/01-sizing.md` (LSK-CAL-001), `docs/04-calcs/sizing.py` , `docs/04-calcs/results.csv` .
- Bill of materials: `bom/bom.csv` .
- Pictures: `cad/src/build_plan_media.py` .
- Design changes: `docs/decisions/0002-design-for-construction.md` (LSK-DDR-002).